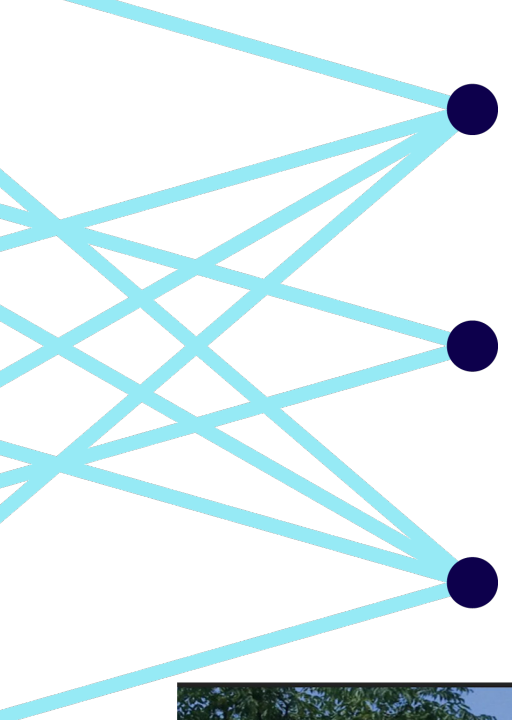




Protein Language Models and Using them for Downstream Tasks

Oznur Tastan, PhD

Zeynep Işık & Mert Pekey

Sabanci University, Istanbul, Turkey

Machine Learning for Multiple Domains: From
Concepts to Implementation

Oct 14, 2024

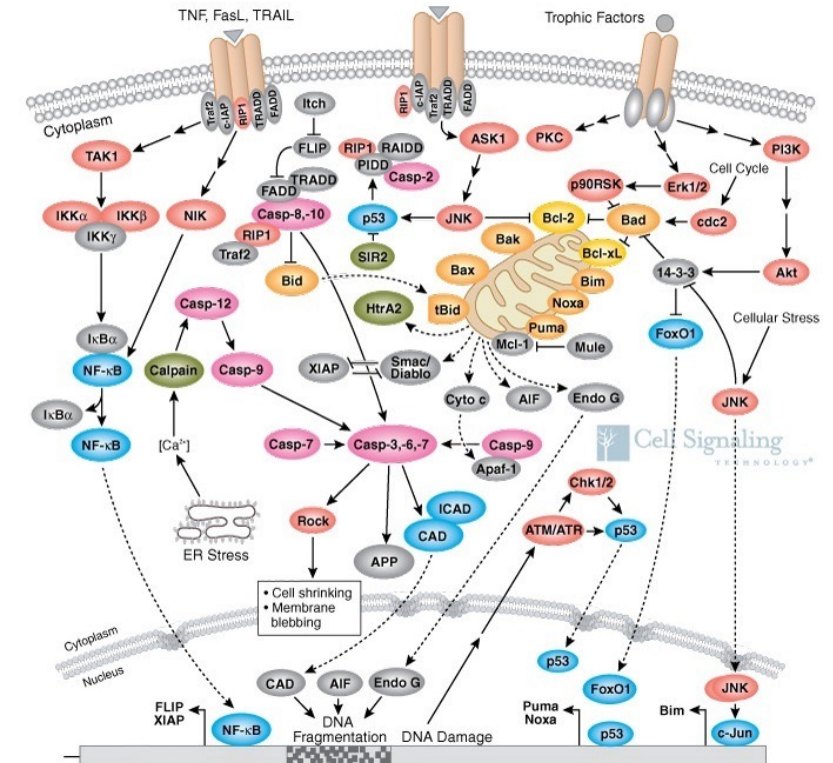


Sabancı
Universitesi



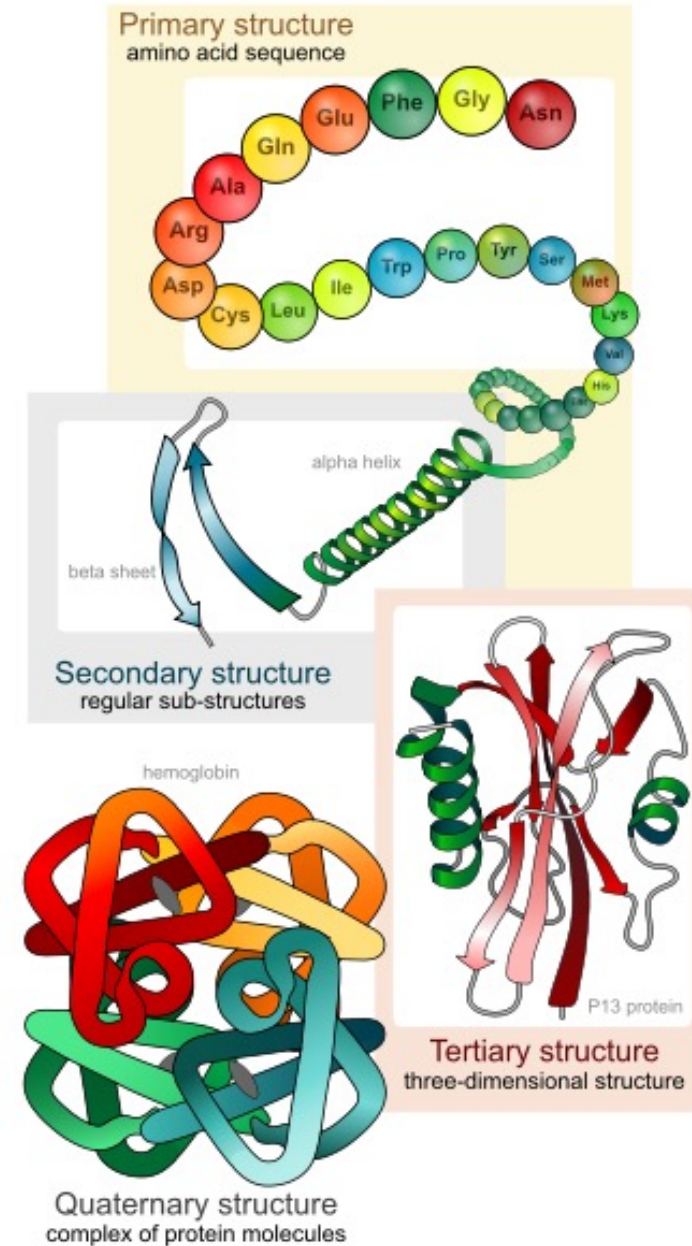
Proteins

- **Proteins** are instrumental in about everything in the cell
 - structural support
 - transport of other substances
 - intercellular signaling
 - movement
 - defense against foreign substances
 - proteins are the main enzymes in a cell and regulate metabolism by selectively accelerating chemical reactions.
- A human cell has more than 20.000 different proteins, each with their structure and functions



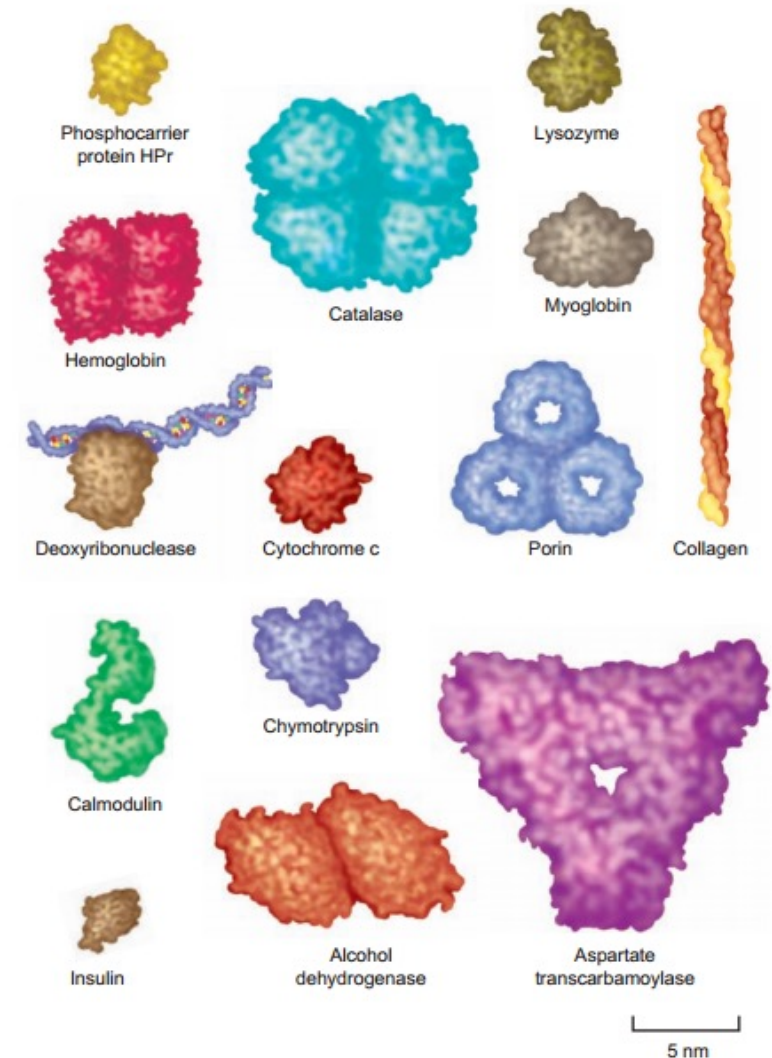
http://www.cellsignal.com/reference/pathway/Apoptosis_Overview.html

Protein Sequences



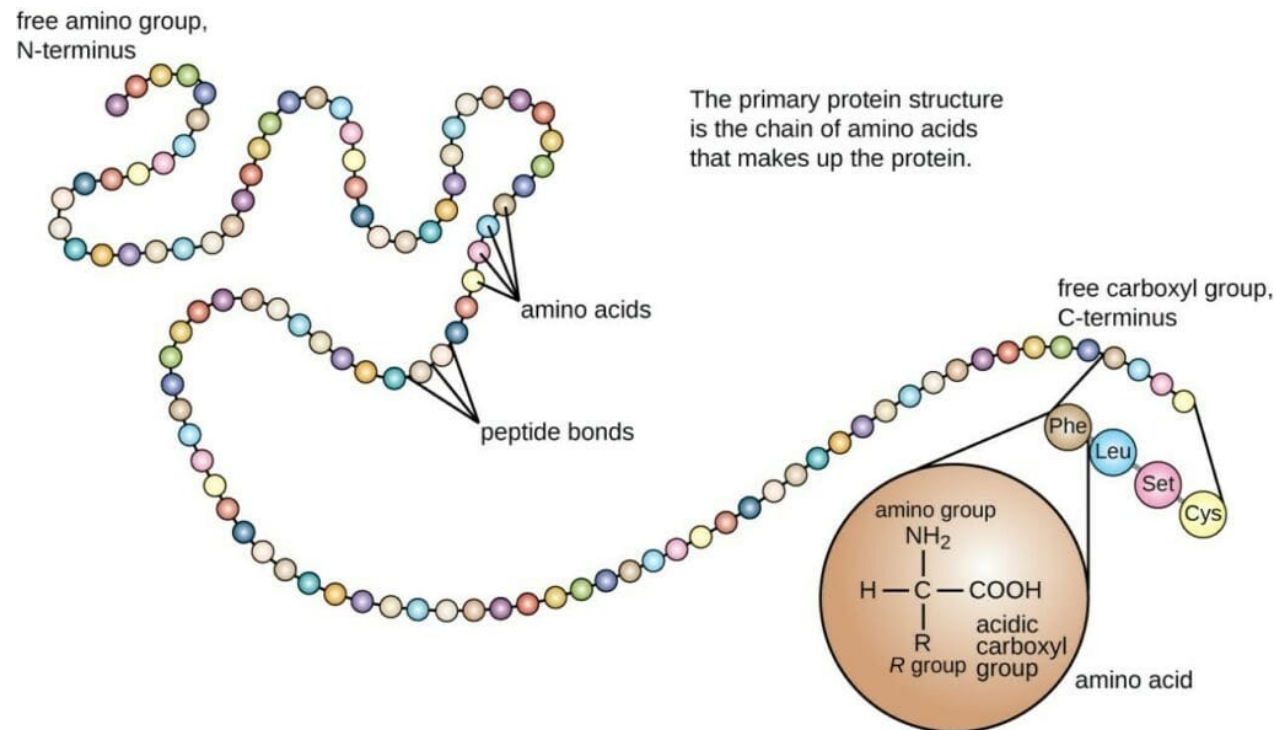
Proteins are diverse

- The protein sequence space is vast
- Subset structure & function is the result of evolutionary pressure
 - Structure features facilitating folding
 - Presence of binding site
 - Structural flexibility/rigidity
 - Appropriate surface structure



Primary sequence

```
>sp|P68871|HBB_HUMAN Hemoglobin subunit beta OS=Homo sapiens OX=9606 GN=HBB PE=1 SV=2
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAVMGNPK
VKAHGKKVLGAFSDGLAHLNKLKGTFAITLSEIHCDKLHVDPENFRLLGNVLVCVLAHHFG
KEFTPPVQAAYQKVVAGVANALAHKYH
```



Amino acids

Amino Acid	Three-Letter Abbreviation	One-Letter Abbreviation
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic acid	Asp	D
Cysteine	Cys	C
Glutamic acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

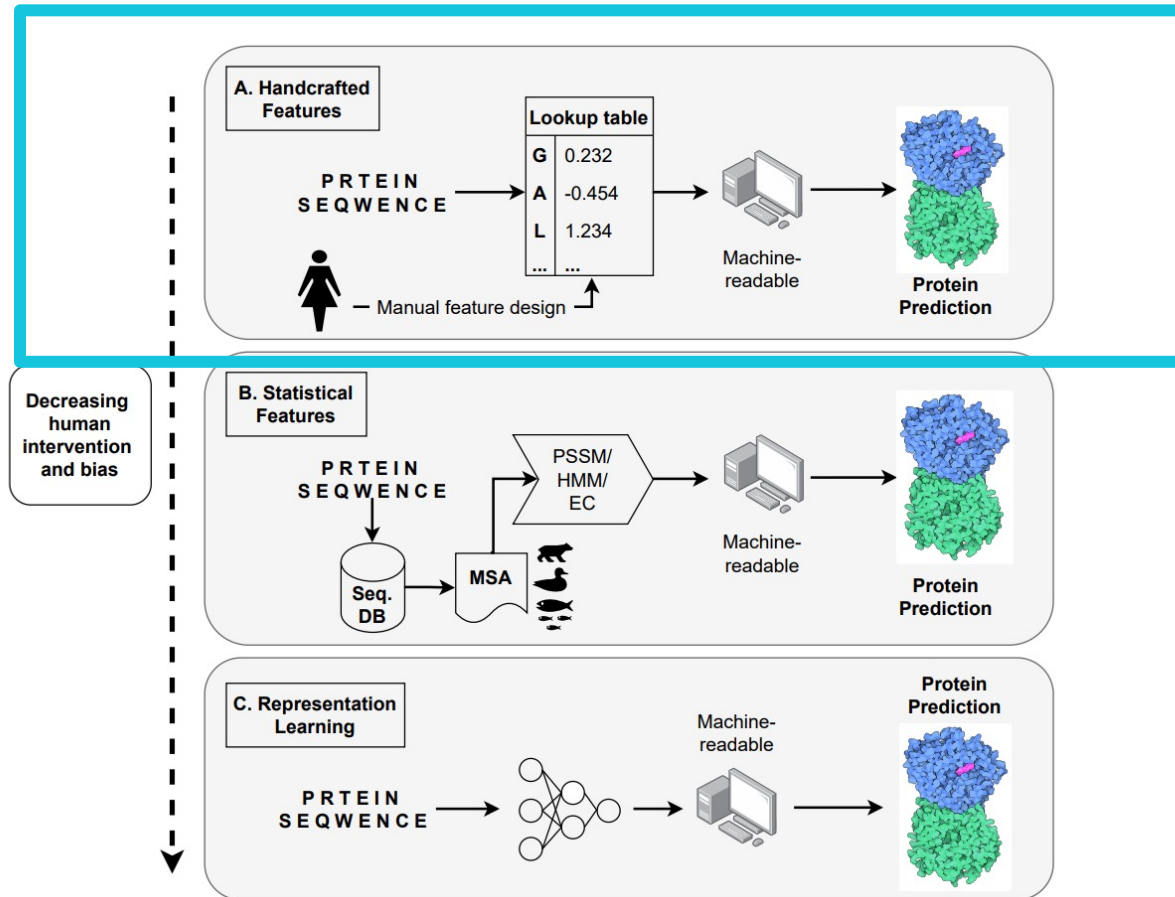
Protein prediction tasks

```
>sp|P68871|HBB_HUMAN Hemoglobin subunit beta OS=Homo sapiens OX=9606 GN=HBB PE=1 SV=2
MVHLTPEEKSAVTALWGKVNVDENVGGEALGRLLVVYPWTQRFFESFGDLSTPDAMGNPK
VKAHGKKVLGAFSDGLAHLNKLKGTFTLSELHCDKLHVDPENFRLLGNVLVCVLAHHFG
KEFTPPVQAAYQKVVAGVANALAHKYH
```

- Given a protein's sequence? **Predict:**
 - Function
 - Subcellular localization
 - Structure prediction (Secondary structure, fold, tertiary)
 - Post-translational modifications the protein could undergo
 - Effect of mutation on foldability, stability, function of the protein
 - Binding to drugs
 - Binding to other molecules (protein, DNA, RNA, ligands)

PART 1 – How did we use to represent protein sequences?

The representation of the protein sequence



Representations – One Hot Encoding

One-Hot Representation

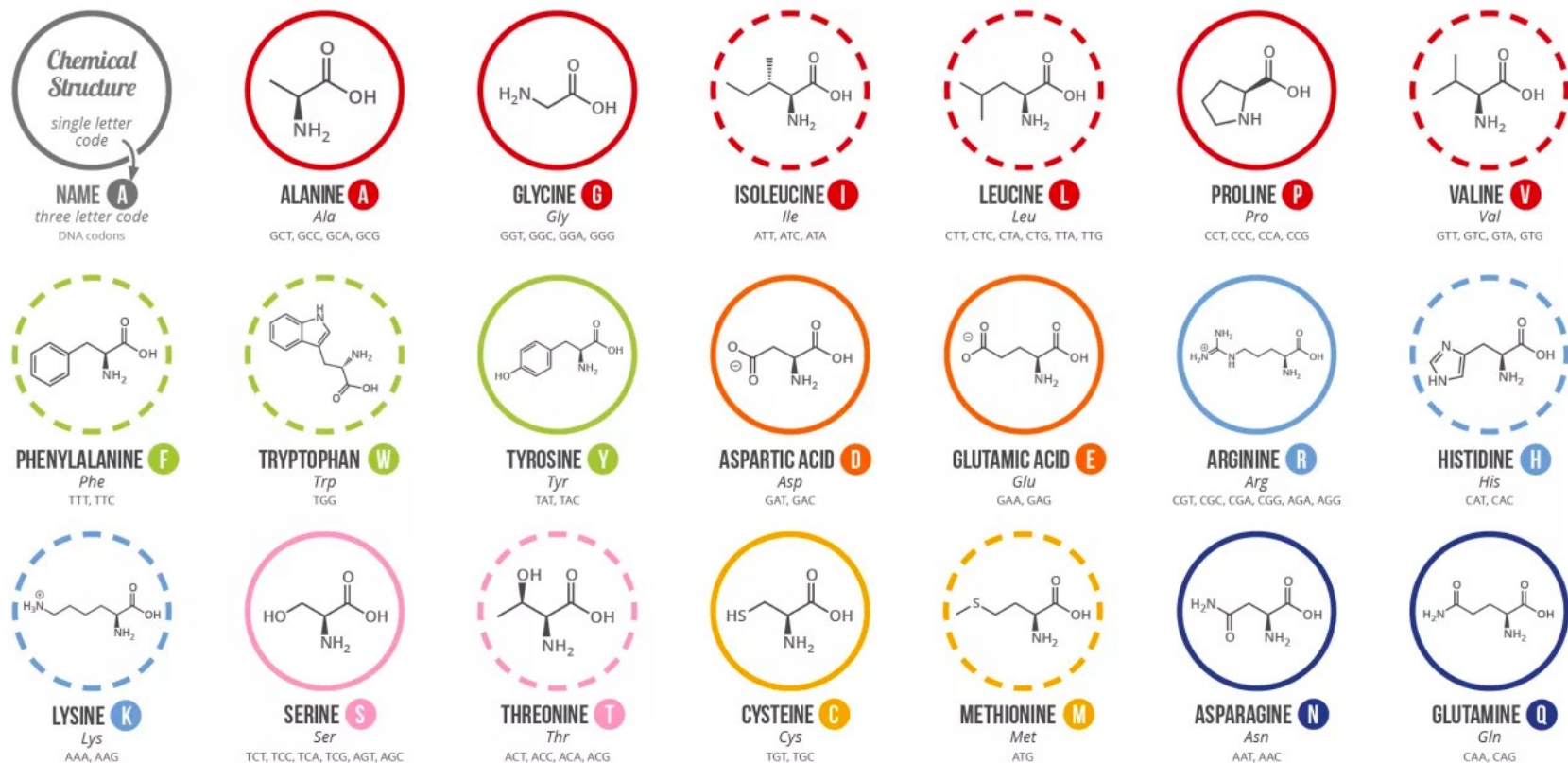
	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	X	Y
K	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0

Does not involve any “semantic” or physicochemical relationships among the amino acid types

A GUIDE TO THE TWENTY COMMON AMINO ACIDS

AMINO ACIDS ARE THE BUILDING BLOCKS OF PROTEINS IN LIVING ORGANISMS. THERE ARE OVER 500 AMINO ACIDS FOUND IN NATURE - HOWEVER, THE HUMAN GENETIC CODE ONLY DIRECTLY ENCODES 20. 'ESSENTIAL' AMINO ACIDS MUST BE OBTAINED FROM THE DIET, WHILST NON-ESSENTIAL AMINO ACIDS CAN BE SYNTHESISED IN THE BODY.

Chart Key: ● ALIPHATIC ● AROMATIC ● ACIDIC ● BASIC ● HYDROXYLIC ● SULFUR-CONTAINING ● AMIDIC ○ NON-ESSENTIAL ○ ESSENTIAL



Note: This chart only shows those amino acids for which the human genetic code directly codes for. Selenocysteine is often referred to as the 21st amino acid, but is encoded in a special manner. In some cases, distinguishing between asparagine/aspartic acid and glutamine/glutamic acid is difficult. In these cases, the codes asx (B) and glx (Z) are respectively used.

Amino acid properties

- ☐ Molecular weight
- ☒ Polarity / Grantham
- ☐ Hphob. OMH / Sweet et al.
- ☐ Hphob. / Abraham & Leo
- ☐ Hphob. / Guy
- ☐ Hphob. / Roseman
- ☐ Hphob. HPLC / Wilson & al
- ☐ Hphob. / Rf mobility
- ☐ HPLC / retention pH 2.1
- ☐ Hphob. / Chothia
- ☐ Average flexibility
- ☐ alpha-helix / Deleage & Roux
- ☐ alpha-helix / Levitt
- ☐ Antiparallel beta-strand
- ☐ Relative mutability
- ☐ Number of codon(s)
- ☐ Refractivity
- ☐ Hphob. / Hopp & Woods
- ☐ Hphob. / Black
- ☐ Hphob. / Janin
- ☐ Hphob. / Tanford
- ☐ Hphob. HPLC / Parker & al
- ☐ HPLC / HFBA retention
- ☐ HPLC / retention pH 7.4
- ☐ Hphob. / Rose & al
- ☐ alpha-helix / Chou & Fasman
- ☐ beta-sheet / Deleage & Roux
- ☐ beta-sheet / Levitt
- ☐ Parallel beta-strand
- ☐ Bulkiness
- ☐ Recognition factors
- ☐ Hydropath. / Kyte & Doolittle
- ☐ Hphob. / Bull & Breese
- ☐ Hphob. / Miyazawa et al.
- ☐ Hphob. / Wolfenden et al.
- ☐ Hphob. HPLC pH3.4 / Cowan
- ☐ HPLC / TFA retention
- ☐ % buried residues
- ☐ Ratio hetero end/side
- ☐ beta-sheet / Chou & Fasman
- ☐ beta-turn / Deleage & Roux
- ☐ beta-turn / Levitt
- ☐ A.A. composition
- ☐ Polarity / Zimmerman
- ☐ Hphob. / Eisenberg et al.
- ☐ Hphob. / Manavalan et al.
- ☐ Hphob. / Fauchere et al.
- ☐ Hphob. / Rao & Argos
- ☐ Hphob. / Welling & al
- ☐ Hphob. HPLC pH7.5 / Cowan
- ☐ Transmembrane tendency
- ☐ % accessible residues
- ☐ Average area buried
- ☐ beta-turn / Chou & Fasman
- ☐ Coil / Deleage & Roux
- ☐ Total beta-strand
- ☐ A.A. comp. in Swiss-Prot

Encode the aminoacids with precomputed numbers

Polarity

Reference: Science 185:862-864(1974).

Amino acid scale values:

Ala: 8.100
Arg: 10.500
Asn: 11.600
Asp: 13.000
Cys: 5.500
Gln: 10.500
Glu: 12.300
Gly: 9.000
His: 10.400
Ile: 5.200
Leu: 4.900
Lys: 11.300
Met: 5.700
Phe: 5.200
Pro: 8.000
Ser: 9.200
Thr: 8.600
Trp: 5.400
Tyr: 6.200
Val: 5.900

Hydrophobicity

Author(s):

Kyte J., Doolittle R.F.

Reference: J. Mol. Biol. 157:105-132(1982).

Amino acid scale values:

Ala: 1.800
Arg: -4.500
Asn: -3.500
Asp: -3.500
Cys: 2.500
Gln: -3.500
Glu: -3.500
Gly: -0.400
His: -3.200
Ile: 4.500
Leu: 3.800
Lys: -3.900
Met: 1.900
Phe: 2.800
Pro: -1.600
Ser: -0.800
Thr: -0.700
Trp: -0.900
Tyr: -1.300
Val: 4.200

Representations – Nlf Encoding

Nlf Representation

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
K	2	0.26	1.2	0.98	0.05	0.32	0.1	0.73	0.11	0.19	0.14	0.02	0.03	0.02	0.05	0.01	0.01

Table 2
Eighteen NLF factors for amino acids.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
A	0.42	-2.07	-0.67	0.01	-1.10	-0.32	-0.20	0.09	-0.20	0.09	-0.11	0.15	0.01	0.06	0.02	0.16	0.07	-0.03
R	1.65	1.40	-0.01	-0.88	-0.08	-0.07	0.60	-0.53	-0.10	0.01	0.09	-0.07	0.09	0.08	0.03	0.09	0.03	-0.02
N	1.68	0.30	-0.49	0.15	0.09	0.59	-0.06	0.02	0.14	0.00	-0.14	-0.09	0.08	-0.14	-0.11	-0.01	0.01	0.01
D	0.81	0.13	1.36	0.63	-0.15	-0.10	-0.45	-0.31	-0.10	0.03	0.15	0.02	0.16	0.12	-0.07	-0.11	-0.01	-0.05
C	-2.70	-0.32	1.19	1.37	0.04	-0.18	0.64	0.21	0.26	0.35	-0.02	-0.11	0.05	-0.04	-0.03	0.10	0.04	-0.04
Q	1.71	1.11	-0.08	0.15	0.11	0.45	0.11	0.08	0.02	0.25	-0.12	0.25	-0.20	0.16	-0.01	-0.07	0.02	0.03
E	1.56	0.48	0.87	-0.02	-0.07	0.13	-0.22	-0.15	-0.09	0.10	0.04	0.05	-0.12	-0.28	0.03	0.09	-0.06	0.02
G	1.32	-2.05	-0.60	0.31	0.61	-0.58	0.00	-0.30	0.44	-0.14	0.18	0.12	-0.12	0.01	0.06	0.00	-0.04	0.01
H	0.13	1.50	-1.22	0.52	-1.14	-0.45	0.13	0.04	0.10	-0.07	0.11	-0.13	-0.06	-0.07	-0.01	-0.16	-0.06	0.03
I	-1.52	-0.45	-0.39	-0.36	-0.01	0.55	0.06	0.10	-0.02	0.08	0.10	0.12	0.18	0.01	0.12	-0.02	-0.20	-0.01
L	-1.29	-1.21	-0.25	-0.96	0.18	0.06	-0.04	0.00	-0.09	0.26	0.18	-0.05	0.00	-0.11	0.01	-0.15	0.14	0.02
K	2.03	0.26	1.22	-0.98	-0.05	-0.32	0.10	0.73	0.11	-0.19	0.14	0.02	0.03	0.02	-0.05	0.01	-0.01	-0.02
M	-1.72	0.85	-0.34	0.44	-0.01	0.80	-0.16	0.05	0.05	-0.30	0.29	0.06	-0.02	0.03	0.03	0.09	0.14	0.00
F	-2.37	0.23	-0.09	-0.37	0.19	-0.04	0.03	-0.06	-0.14	-0.14	-0.10	0.03	-0.21	-0.04	-0.09	-0.03	-0.06	-0.18
P	1.41	0.27	-1.09	0.77	0.87	-0.33	-0.04	0.27	-0.43	0.06	0.10	-0.14	0.03	0.03	-0.01	0.04	-0.03	0.01
S	1.47	-1.11	-0.27	0.13	0.15	0.22	0.09	-0.05	0.05	-0.14	-0.30	0.01	0.16	-0.03	-0.01	-0.06	0.05	-0.06
T	0.30	-0.68	0.88	0.23	-0.10	0.23	0.03	-0.01	-0.14	-0.16	-0.15	-0.20	-0.12	0.05	0.21	-0.07	0.00	0.04
W	-2.83	1.79	0.16	-0.14	0.42	-0.84	-0.13	-0.06	-0.04	-0.18	-0.32	0.26	0.15	-0.08	0.04	-0.01	0.06	0.12
Y	-0.70	0.95	-0.36	-0.60	0.09	-0.06	-0.55	0.01	0.28	0.17	-0.12	-0.23	-0.02	0.13	0.05	0.08	-0.02	-0.02
V	-1.33	-1.39	0.15	-0.40	-0.04	0.27	0.07	-0.12	-0.10	-0.06	-0.01	-0.09	-0.07	0.10	-0.20	0.02	-0.08	0.13

Nanni and Lumini, “A new encoding technique for peptide classification,” Expert Syst. Appl., vol. 38, no. 4, pp. 3185–3191, 2011.

Reduced Alphabet

AA	Charge	Polarity	Aromaticity	Size	Electronic Property
A	Neutral	Non-polar	Neutral	Small	Strong Donor
R	Positive	Polar	Neutral	Large	Strong Acceptor
N	Neutral	Polar	Neutral	Medium	Strong Acceptor
D	Negative	Polar	Neutral	Medium	Strong Donor
C	Neutral	Polar	Neutral	Large	Neutral
Q	Neutral	Polar	Neutral	Large	Weak Acceptor
E	Negative	Polar	Neutral	Large	Strong Donor
G	Neutral	Non-polar	Neutral	Small	Neutral
H	Positive	Polar	Aromatic	Large	Neutral
I	Neutral	Non-polar	Aliphatic	Large	Weak Donor
L	Neutral	Non-polar	Aliphatic	Large	Weak Donor
K	Positive	Polar	Neutral	Large	Strong Acceptor
M	Neutral	Non-polar	Neutral	Large	Weak Acceptor
F	Neutral	Non-polar	Aromatic	Large	Weak Acceptor
P	Neutral	Non-polar	Neutral	Small	Strong Donor
S	Neutral	Polar	Neutral	Small	Neutral
T	Neutral	Polar	Neutral	Medium	Weak Acceptor
W	Neutral	Non-polar	Aromatic	Large	Neutral
Y	Neutral	Polar	Aromatic	Large	Weak Acceptor
V	Neutral	Non-polar	Aliphatic	Large	Weak Donor

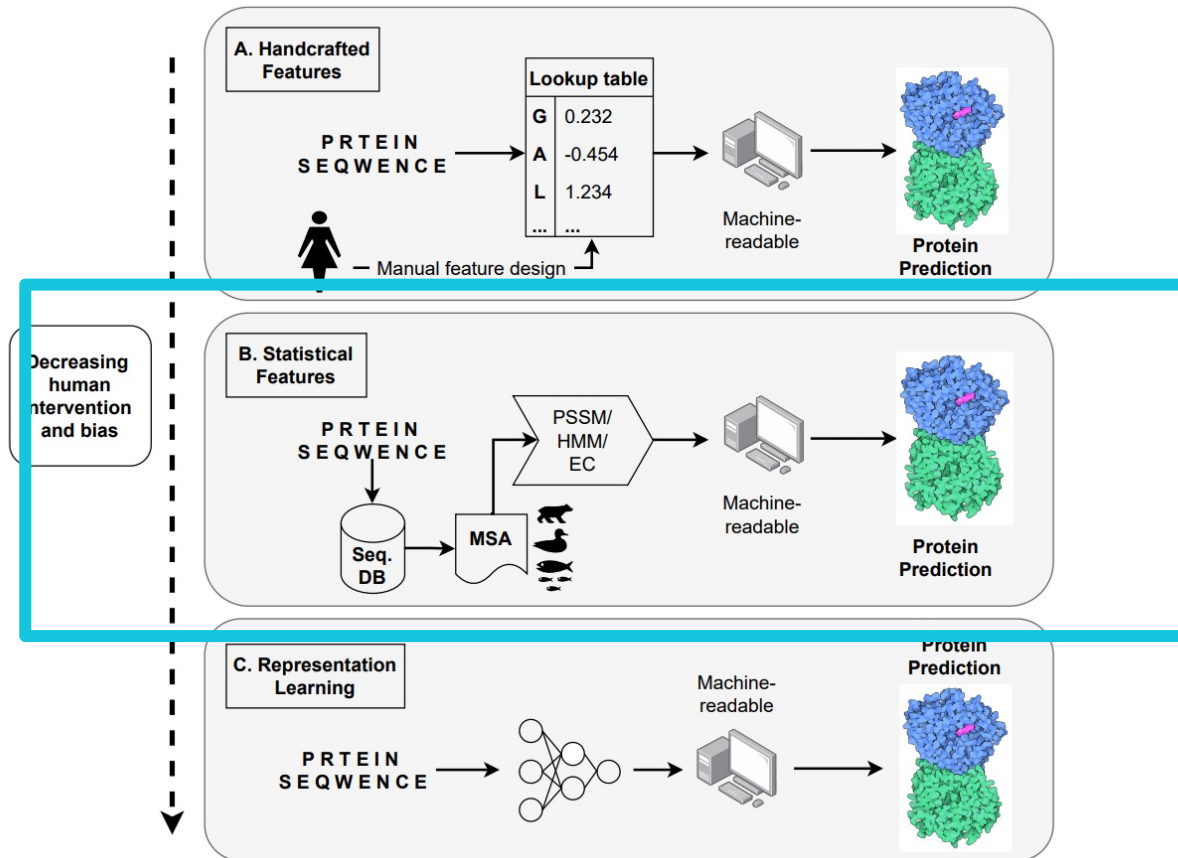
BLOSUM encoding

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V	B	Z	X	*
K	-1	2	0	-1	-3	1	1	-2	-1	-3	-2	5	-1	-3	-1	0	-1	-3	-2	-2	0	1	-1	-4

BLOSUM62 Substitution Matrix

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V	B	Z	X	*
A	4	-1	-2	-2	0	-1	-1	0	-2	-1	-1	-1	-1	-2	-1	1	0	-3	-2	0	-2	-1	0	-4
R	-1	5	0	-2	-3	1	0	-2	0	-3	-2	2	-1	-3	-2	-1	-1	-3	-2	-3	-1	0	-1	-4
N	-2	0	6	1	-3	0	0	0	1	-3	-3	0	-2	-3	-2	1	0	-4	-2	-3	3	0	-1	-4
D	-2	-2	1	6	-3	0	2	-1	-1	-3	-4	-1	-3	-3	-1	0	-1	-4	-3	-3	4	1	-1	-4
C	0	-3	-3	-3	9	-3	-4	-3	-3	-1	-1	-3	-1	-2	-3	-1	-1	-2	-2	-1	-3	-3	-2	-4
Q	-1	1	0	0	-3	5	2	-2	0	-3	-2	1	0	-3	-1	0	-1	-2	-1	-2	0	3	-1	-4
E	-1	0	0	2	-4	2	5	-2	0	-3	-3	1	-2	-3	-1	0	-1	-3	-2	-2	1	4	-1	-4
G	0	-2	0	-1	-3	-2	-2	6	-2	-4	-4	-2	-3	-3	-2	0	-2	-2	-3	-3	-1	-2	-1	-4
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L	-1	-2	-3	-4	-1	-2	-3	-4	-3	2	4	-2	2	0	-3	-2	-1	-2	-1	1	-4	-3	-1	-4
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F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6	-4	-2	-2	1	3	-1	-3	-3	-1	-4
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S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4	1	-3	-2	-2	0	0	0	-4
T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5	-2	-2	0	-1	-1	0	-4
W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-1	1	-4	-3	-2	11	2	-3	-4	-3	-2	-4
Y	-2	-2	-2	-3	-2	-1	-2	-3	2	-1	-1	-2	-1	3	-3	-2	-2	2	7	-1	-3	-2	-1	-4
V	0	-3	-3	-3	-1	-2	-2	-3	-3	3	1	-2	1	-1	-2	-2	0	-3	-1	4	-3	-2	-1	-4
B	-2	-1	3	4	-3	0	1	-1	0	-3	-4	0	-3	-3	-2	0	-1	-4	-3	-3	4	1	-1	-4
Z	-1	0	0	1	-3	3	4	-2	0	-3	-3	1	-1	-3	-1	0	-1	-3	-2	-2	1	4	-1	-4
X	0	-1	-1	-1	-2	-1	-1	-1	-1	-1	-1	-1	-1	-1	-2	0	0	-2	-1	-1	-1	-1	-1	-4
*	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	1

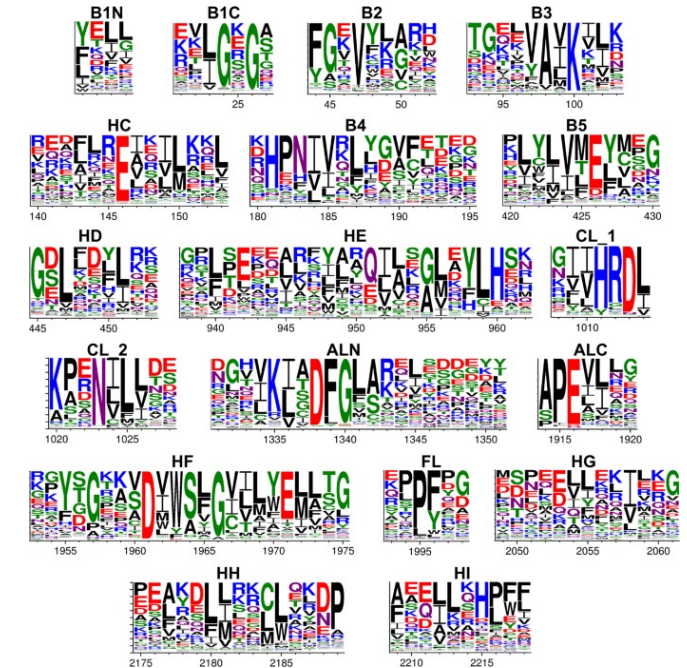
The representation of the protein sequence



Features on Multiple Sequence Alignment

```

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A7S3R2_NEMVE/58-276        .....SLSKDAMKLFHFKMEGVSNGHCFEIQQVGEGKAFDGEHWSKLCVVKGKHLPFSDILMPMSYGTQKQFAPYPA
C3ZLQ0_BRAFL/370-580       ..msvptn-----LDLHIYGSINGMEFDMVGGGSGNPNDGSLSVNVKSTKGA-LRVSPLLVGPHLGYGHYQYLPFPD
A7S1H3_NEMVE/1-82          .....-----MFGSKAFAPYPD
A7RGQ5_NEMVE/1-174         .....-----MKLHFKEGVSNGHCFEIQQVGEGKPFEGEQWAKHCVVKGKHLPFSLDIIMPNI-----TFAPYPD
C3YKP7_BRAFL/5-213         ....athe-----IHLHGSVNGHEFDLVGSGKGDPPKAGSLVTEVKSTMGK-LKFSPLHMPHLGYGYQYLPYPD
U6M5D0_EIMMA/34-255        .....ELFTGVVPILVELDGDVNGHKFSVSGEGGDATYGKCLKICTGK-LPVPWPTLVTTFGYGLMCFARYPD
C3YR99_BRAFL/49-259        ...ptthe-----LHIFGSFNGVEFDMVGRGIGNPNDGYEELNLKSTKGA-LKFSPLWLVFQIGYGFHGYLPYPD
W2JZJ6_PHYPR/96-317        .....ELFTGIVPILIELNGDVNGHKFSVSGEGGDATYGKLTCLKICTGK-LPVPWPTLVTTLSYGVQCFARYPD
C3YRA0_BRAFL/13-103        ep1ptthe-----LHIFGSFNGVEFDLVGRGEGNPKDGSQNLHLKSTKGA-LQSPWMLVPHIGYGFYQYLPYPD
U6GSR1_EIMAC/7-228        .....ELFTGVVPILVELDGDVNGHKFSVSGEGGDATYGKLTCLKICTGK-LPVPWPTLVTTFGYGLMCFARYPD
C3YKI0_BRAFL/3-212        ..nksvpt-----NLDLHIYGSINGMEFDMVGGGSGNPNDGSLAVNVKSTKGA-LRVSPLLVGPHLGYGHYQYLPFPD
C3YRA2_BRAFL/11-221        ...pkthe-----LHIFGSFNGVEFDMVGRGIGNPNEGSEELNAKFTKGP-LKFSPLYLVPHLGYAYYQYLPFPD
C3YKQ3_BRAFL/5-214         ....athd-----IHLHGSINGHEFDLVGSGKGDPPKAGSLVTTAKSTKGA-LKFSPLYLMPHLGYGYQYLPYPD
A0A2B4RJ70_STYPI/16-192    .....e-IIQDDMKMEYEMKGVNGHEFTIEGEGNGKPYEGKQTANFKVITGAPLSFSDIPSSVFQYGNRCFTYPE
C3YRA1_BRAFL/52-263        ...pkthe-----LHIFGSFNGVGFDMVGEGTGNPNEGSEELKLKSTNGR-LKFSPLYLVPHLGYAFNQYLPFPD
C3YKP9_BRAFL/5-212         ....tahd-----LHIFGSVNGAEFDLVGGGKGNPNDGTLETSVKSTRGA-LPCSPLLIQPNLGYGFYQYLPFPD
A0A1V4T0E4_9GAMM/10-189   .....ELFTGVVPILVELDGDVNGHKFSVSGEGGDATYGKLTCLKICTGK-LPVPWPTLVTTFGYGLQCFARYPD
C3YRA3_BRAFL/5-214         ....tthe-----VHVYGSINGVEFDLVGSGKGNPKDGSSEIQVKSTKGP-LGFSPLYIVVFNIGYGFHGYLPFPD
C3YKQ1_BRAFL/5-213        ....tthd-----LHIFGSVNGAEFDLVGGGKGNPNDGTLETSVKSTRGA-LPCSPLLIQPNLGYGFYQYLPFPD
A0A2B4RBW9_STYPI/44-153   .....npy-----
U6M9S0_EIMMA/1-78         .....q-----
J8VIQ3_9SPHN/6-227        .....ELFTGVVPILVELDGDVNGHKFSVSGEGGDATYGKLTCLKICTGK-LPVPWPTLVTTFGYGLQCFARYPD
A0A2B4R4L6_STYPI/1-109    .....m-----
C3YKQ0_BRAFL/6-195        ....ahdc-----HMFSGSINGHEFDLVGGGNGNPNDGTLETVKVRSKGA-LPFSPLVILAPNLGYGFYQYLPFPD
A7S1H5_NEMVE/2-221        .....SLSKDAMKLFHFKMEGVSNGHCFEIQQVGEGKAFDGEHWSKLCVVKGKHLPFSDILAPCLKYGSKPFVVKYPD
    
```



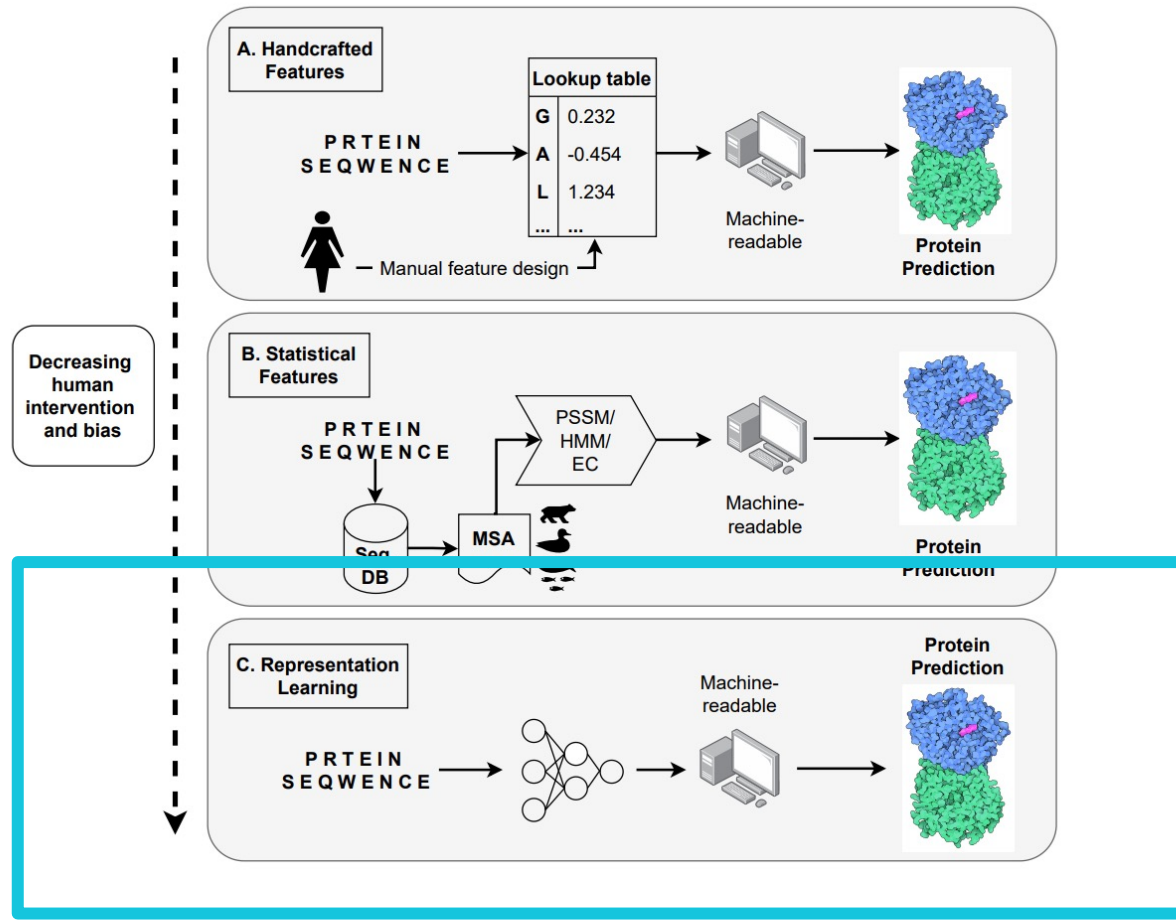
Kinase domain

- Presence/absence of motifs
- Position specific scoring matrices
- Conservation scores
- HMM profiles

Modi et al. Scientific Reports, 2019

PART 2 – How do we **learn** representation of protein sequences?

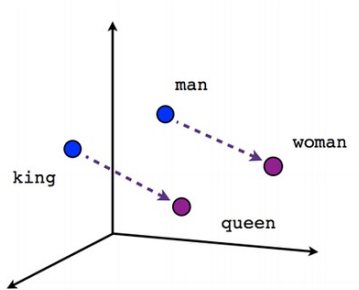
How can we learn the representation?



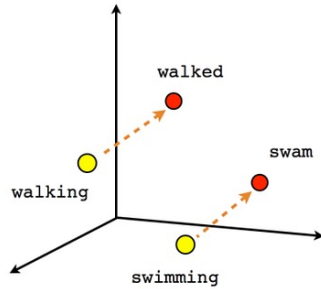
The key idea is to learn representations by training on large data sets of **unlabeled** protein sequences.

ProtVec analogous to Word2Vec

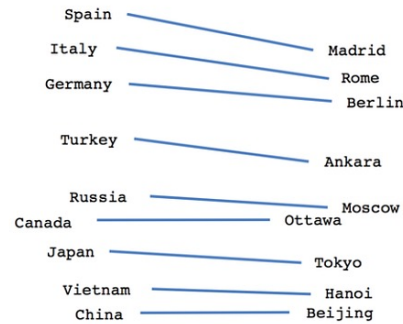
Word2Vec



Male-Female



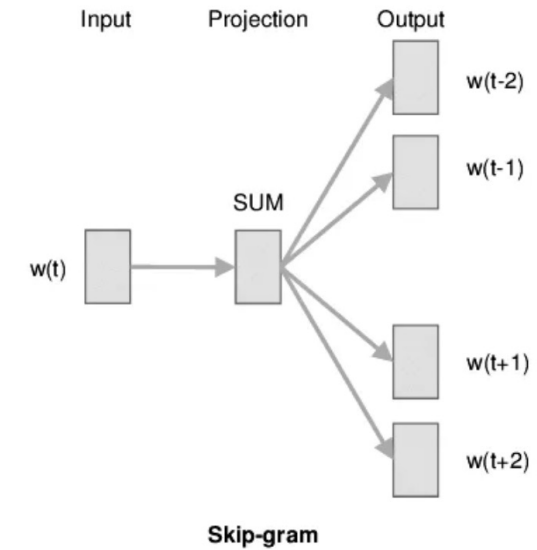
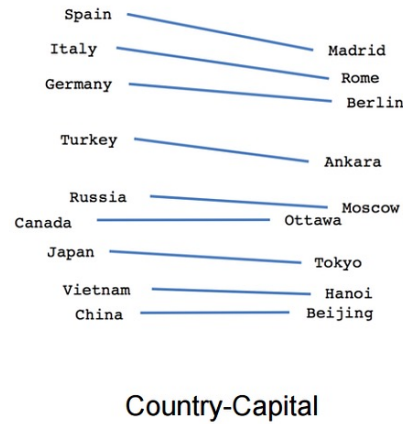
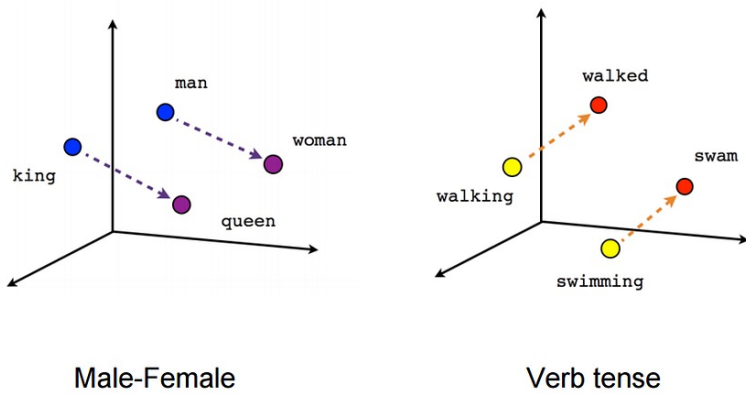
Verb tense



Country-Capital

ProtVec analogous to Word2Vec

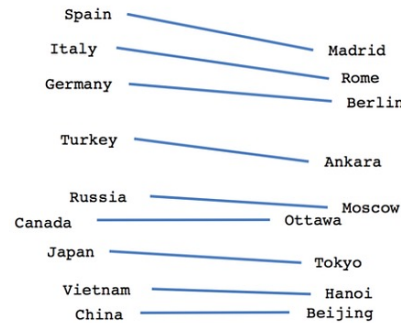
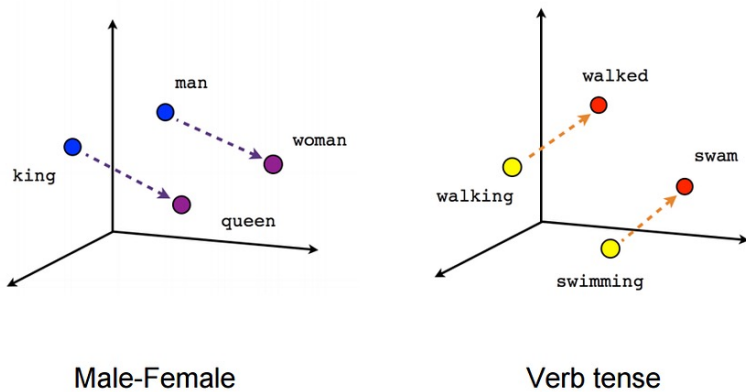
Word2Vec



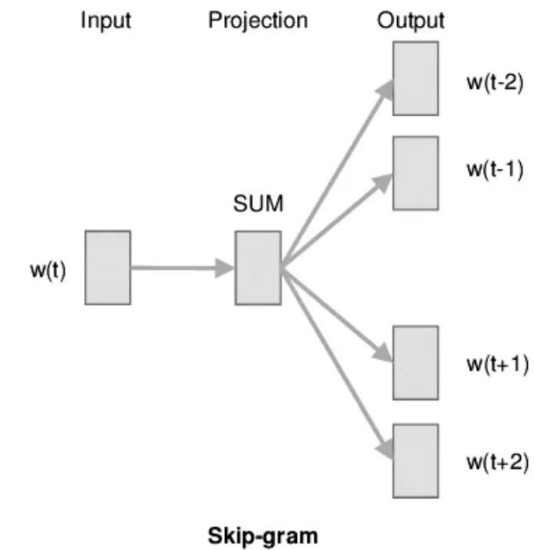
The skip-gram architecture

ProtVec analogous to Word2Vec

Word2Vec



Country-Capital



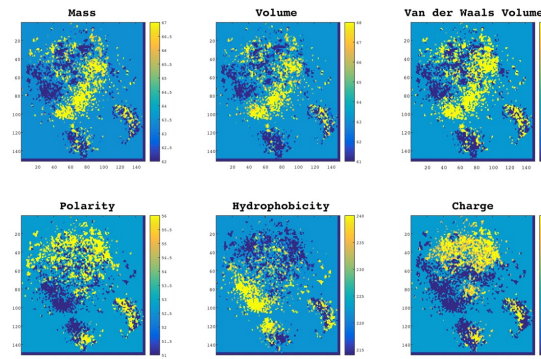
The skip-gram architecture

Original Sequence

(1) $\vec{M}$ (2) $\vec{A}$ (3) $\vec{F}$ SAEDVLKEYDRRRRMEAL..

Splittings

$\left\{ \begin{array}{l} \textcolor{red}{1} \text{ MAF, SAE, DVL, KEY, DRR, RRM, ..} \\ \textcolor{blue}{2} \text{ AFS, AED, VLK, EYD, RRR, RME, ..} \\ \textcolor{green}{3} \text{ FSA, EDV, LKE, YDR, RRR, MEA, ..} \end{array} \right.$



A 100 dimensional vector for each 3-gram

Contextualized embeddings

"The **bug** in the code caused the entire system to crash."

"Suddenly, a **bug** flew into my lemonade, ruining my perfect day."

Same words, different contexts, different semantics.

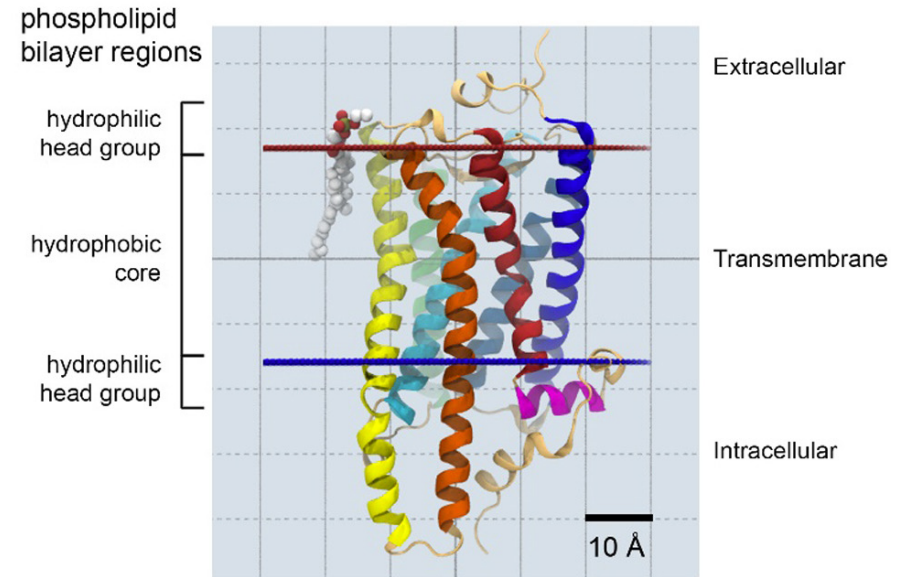
Is this relevant for protein sequences?

Valine (V) in a transmembrane protein:
"...ALVLLVVV..."

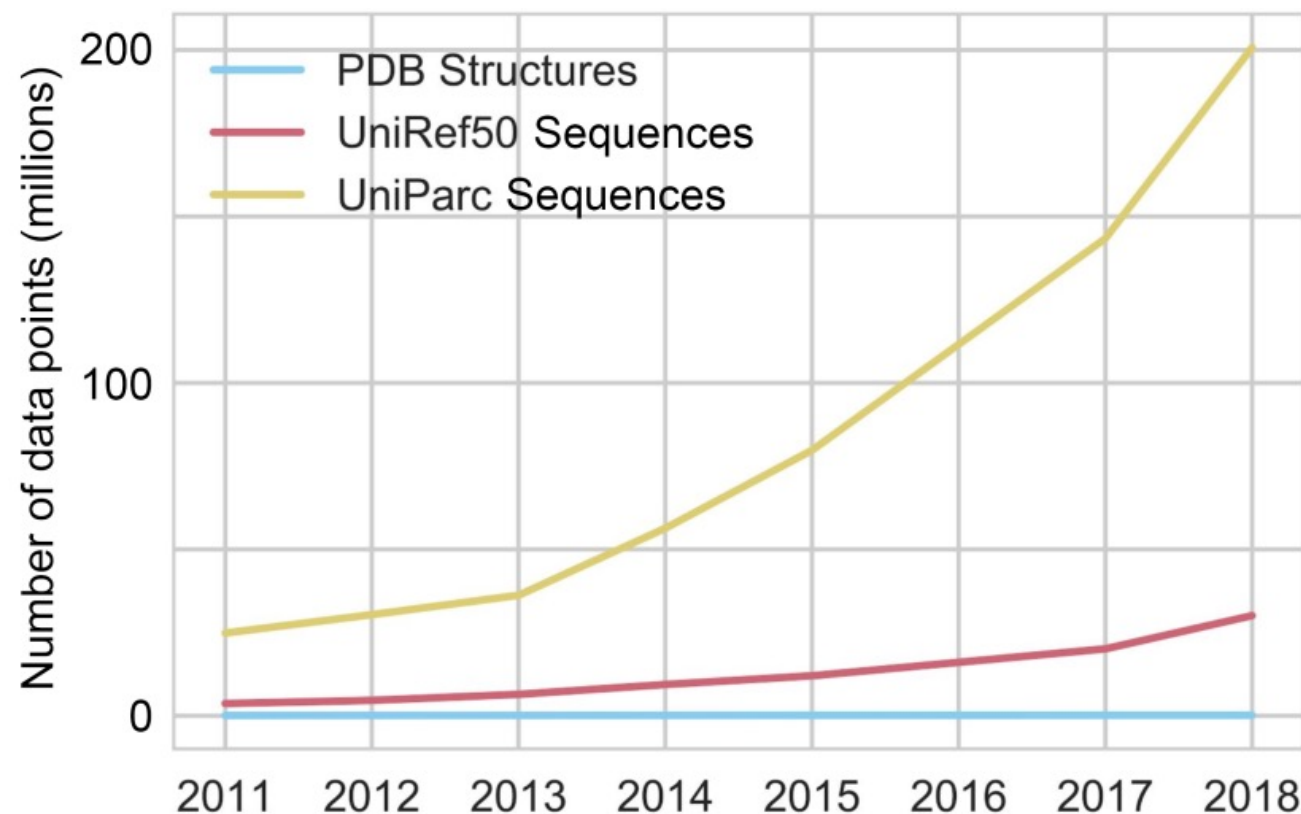
- Here, **valine** might contribute to forming the **hydrophobic core** of a membrane-spanning region, ensuring that the protein integrates properly into the lipid bilayer. In this context, **valine** helps maintain **membrane stability** and **hydrophobicity**.

Valine (V) in a globular protein:
"...FVKQHT..."

- In a water-soluble globular protein, **valine** might appear on the surface, interacting with other hydrophobic patches or even participating in **protein-protein interactions**. In this case, valine's role is more about enabling **intermolecular interactions**.



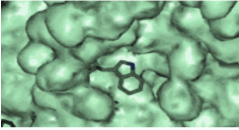
Abundant sequence information



Most pLMs rely on protein sequence data

Because of the sequence-structure knowledge gap

Biological Language Modeling – 2000s



Center For Biological Language Modeling

@

[Carnegie Mellon University](#) | [University of Pittsburgh](#) | [Massachusetts Institute of Technology](#)
[Boston University Medical School](#) | [National Research Council Canada](#)

Research Goals	Publications	BLM Web Toolkit	Conferences	Contact
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New and Past Events

[Biological Language Conference 2005](#) * [New](#)
[Advisory Board Meeting](#)
[Restricted Access](#)
[Biological Language Conference 2004](#)
[NSF VISIT July 2004](#) * [Restricted Access](#)
[Workshop on Ambient Intelligence for Scientific Discovery](#)
[Biological Language Conference 2003](#)
[Biological Language Modeling Workshop May 13 & 14, 2003](#)
[Biological Language Modeling Workshop Nov 7 & 8, 2002](#)

Overview of the Biological Language Modeling Project

The Biological Language Modeling project is based on the assumption that protein sequences from different organisms may be viewed as texts written in different languages. The mapping of protein sequence to their structure, dynamics and function then becomes analogous to the mapping of words to meaning in natural languages. This analogy can be exploited by application of statistical language modeling and text classification techniques to biological sequences, thereby generating testable hypotheses regarding the fundamental building blocks of "protein sequence language". The biology-language analogy enables novel applications of language technologies to the biology domain, but is to a great extent overlapping with existing other computational biology/bioinformatics applications.

Goal 1. Integration of linguistic analyses results and those from structural dynamics for characterizing key residues that control enzymatic functions

Principle Investigators	Ivet Bahar , Michele Loewen , Cathy Costello , Jaime Carbonell , Roni Rosenfeld
Students	Alpay Temiz, Lee-Wei Yang
Postdocs	Dror Tobl , Shyamashi Biswas

- i. Computational assessment of the hinge-bending and other key mechanical sites that coordinate functional dynamics of representative sets of enzymes, using the recently compiled (Thornton and coworkers) enzyme classifications and network models of protein structure and dynamics.
- ii. Systematic analysis of the structural regions/motifs that are implicated in key mechanical roles, using graph theoretical, computer vision and linguistic tools
- iii. Experimental characterization of the mechanism of action of particular classes of enzymes as case studies (e.g. dioxygenase enzymes in collaboration with NRC)
- iv. IMPACT: Discover new potential ligand-binding sites
- v. EVALUATION: Test key mechanical sites predictions against known inhibitor binding sites towards elucidating the possible coupling between mechanical and chemical activities, and hypothesize potential new binding sites to be tested experimentally

Goal 2. Deep understanding of transmembrane protein folding and function

Principle Investigators	Judith Klein-Seetharaman , Gobind Khorana , Ivet Bahar , Raj Reddy , Hagai Meirovitch
Students	Baskar Jain , Madhavi Ganaseetharan , Jianping Miao , Naveena Yanamala
Postdocs	AJ Rader , Harpreet Kaur Dhillon , David Man

- i. Discern property conservation of amino acids (e.g. discriminating hydrophobic amino acids facing outside vs. inside of transmembrane helices)
- ii. Identify the most salient motifs for structural stability and for governing conformational changes, building on our existing results for rhodopsin and leveraging context-dependent statistical grammars or similar approaches.
- iii. Discriminate between functional-relevant and structurally-relevant motifs.
- iv. IMPACT: New strategies to help diagnose and eventually treat conformational diseases associated with transmembrane proteins.
- v. EVALUATION: site-directed mutagenesis experiments.

Goal 3. Deep understanding of relation between b-sheet formation and underlying primary sequences

Principle Investigators	Jaime Carbonell , Jonathan King , Vanathi Gopalakrishnan , Judith Klein-Seetharaman
Students	Yan Liu , Weimin Poos , Ryan Simkovsky
Postdocs	Peter Weigelt

- i. Extension to supersecondary structures via long-range probabilistic linguistic models
- ii. Explore b-sheet transmembrane proteins (incl b-barrels)



Prof. Judith Klein-Seetharaman
Then, University of Pittsburgh
Carnegie Mellon University



Prof. Jaime G. Carbonell
Language Technologies Institute
School of Computer Science
Carnegie Mellon University

<https://www.cs.cmu.edu/~blmt/>

N-gram language model

$$P(S) = \prod_i^N P(s_i | s_{i-(n-1)}, \dots, s_{i-1}) \quad \text{An n-gram model}$$

Before

$P(\text{I saw a cat on a mat}) =$

- $P(\text{I})$
- $\cdot P(\text{saw} | \text{I})$
- $\cdot P(\text{a} | \text{I saw})$
- $\cdot P(\text{cat} | \text{I saw a})$
- $\cdot P(\text{on} | \text{I saw a cat})$
- $\cdot P(\text{a} | \text{I saw a cat on})$
- $\cdot P(\text{mat} | \text{I saw a cat on a})$

After (3-gram)

$P(\text{I saw a cat on a mat}) =$



- | | | |
|---|-------------------|--------------------------------------|
| $P(\text{I})$ | $\longrightarrow$ | $P(\text{I})$ |
| $\cdot P(\text{saw} \text{I})$ | $\longrightarrow$ | $\cdot P(\text{saw} \text{I})$ |
| $\cdot P(\text{a} \text{I saw})$ | $\longrightarrow$ | $\cdot P(\text{a} \text{I saw})$ |
| $\cdot P(\text{cat} \text{I saw a})$ | $\longrightarrow$ | $\cdot P(\text{cat} \text{saw a})$ |
| $\cdot P(\text{on} \text{I saw a cat})$ | $\longrightarrow$ | $\cdot P(\text{on} \text{a cat})$ |
| $\cdot P(\text{a} \text{I saw a cat on})$ | $\longrightarrow$ | $\cdot P(\text{a} \text{cat on})$ |
| $\cdot P(\text{mat} \text{I saw a cat on a})$ | $\longrightarrow$ | $\cdot P(\text{mat} \text{on a})$ |

ignore use

Some historical context



Prof. Madhavi Ganapathiraju

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Document Classification Approach Leads to
a Simple, Accurate, Interpretable
G Protein Coupled Receptor Classifier

Betty Yee Man ChengYMCHENG@CS.CMU.EDU

Jaime G. CarbonellJGC@CS.CMU.EDU

Judith Klein-SeetharamanJUDITHKS@CS.CMU.EDU

Language Technologies Institute
School of Computer Science
Carnegie Mellon University
5000 Forbes Avenue
Pittsburgh, PA 15213, USA

Abstract¹

The need for accurate, automated protein classification methods continues to increase as advances in biotechnology uncovers new proteins at a fast rate. G-protein coupled receptors (GPCRs) are a particularly difficult superfamily of proteins to classify due to the extreme diversity among its members; yet, they are an important subject in pharmacological research being the target of approximately 60% of current drugs (Muller, 2000). A comparison of BLAST, k-NN, HMM and SVM with alignment-based features by Karchin *et al.* (2002) has suggested that classifiers at the complexity of SVM are needed to attain high accuracy in GPCR subfamily classification. Here, analogous to document classification, we applied Decision Tree and Naïve Bayes classifiers with chi-square feature selection on n-gram counts to the GPCR family and subfamily classification task. Using the dataset and evaluation protocol from the previous study, we found the Naïve Bayes classifier surpassing the reported accuracy of SVM by 4.8% and 6.1% in level I and II subfamily classification with an accuracy of 93.2% and 92.4% respectively. The Decision Tree, while inferior to SVM, still outperforms HMM in both level I and II subfamily classification. Moreover, the n-grams selected by chi-square feature selection show evidence of biological importance. Thus, the document classification approach has resulted in a simpler, more accurate and interpretable classifier.

Keywords: GPCR, protein family classification, chi-square, Naïve Bayes, decision tree

Decision Tree		Naïve Bayes	
N-grams Used	Accuracy	N-grams Used	Accuracy
1-gram	89.4 %	2-grams	80.7 %
1, 2-grams	89.5 %	2, 3-grams	96.3 %
1, 2, 3-grams	89.3 %	2, 3, 4-grams	95.6 %
		2, 3, 4, 5-grams	94.8 %

Table 5. Result of ten-fold cross-validation on GPCR classification at the family level using decision tree and Naïve Bayes classifier on all n-grams of the specified sizes.

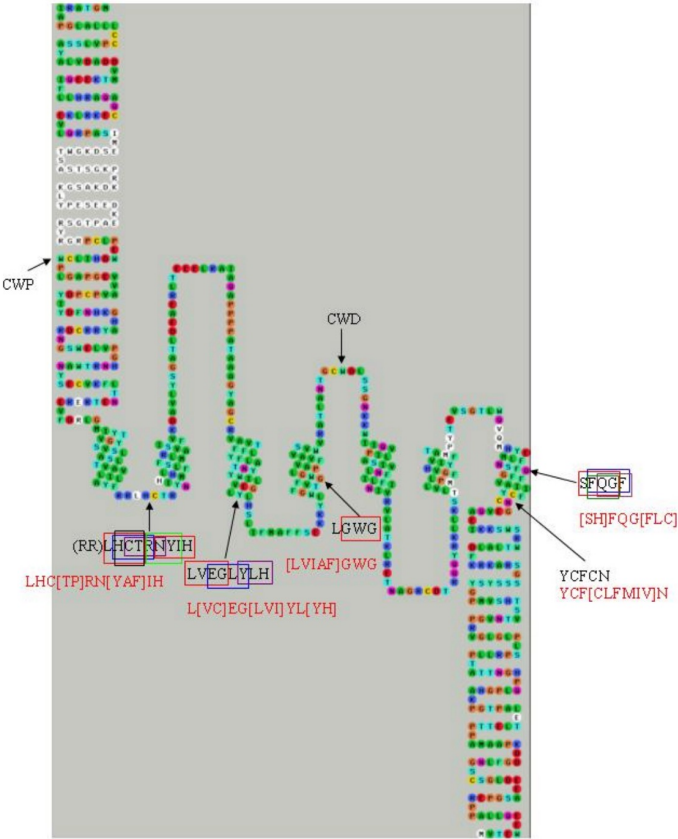
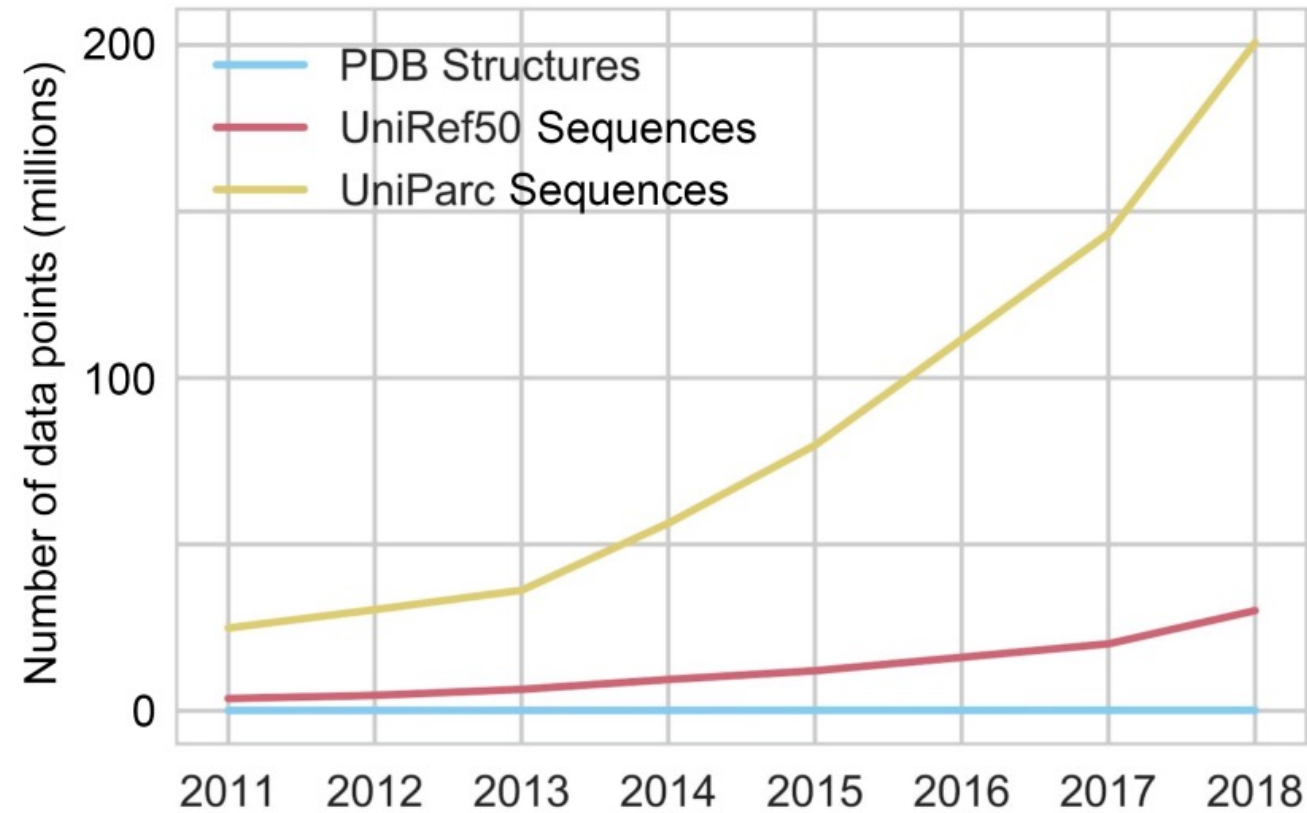


Figure 1. Snake-plot of PTRR_HUMAN in Class B GPCRs with the 20 most significant n-grams as selected by modified chi-square feature selection labeled.

Abundant sequence information



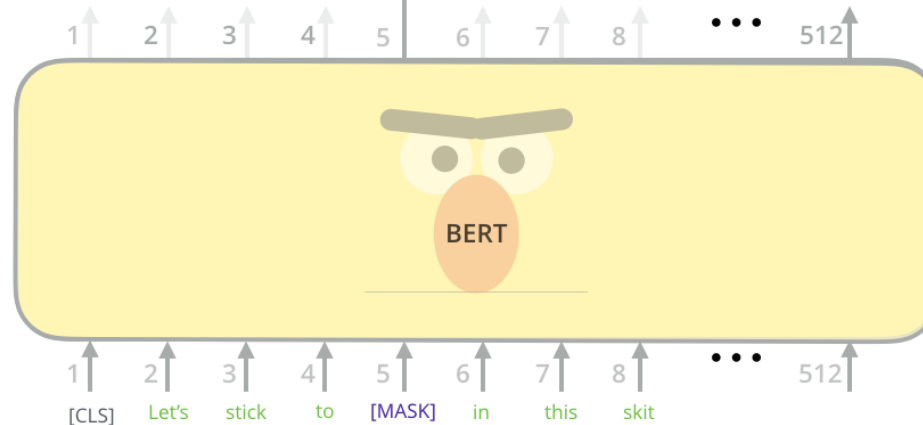
Self-supervised learning

Use the output of the masked word's position to predict the masked word

Possible classes:
All English words

0.1%	Aardvark
...	...
10%	Improvisation
...	...
0%	Zyzzzyva

FFNN + Softmax



Randomly mask
15% of tokens

Input

[CLS] Let's stick to improvisation in this skit

Model implicitly learns syntax, semantics, context .. etc.

Modern Protein Language Models

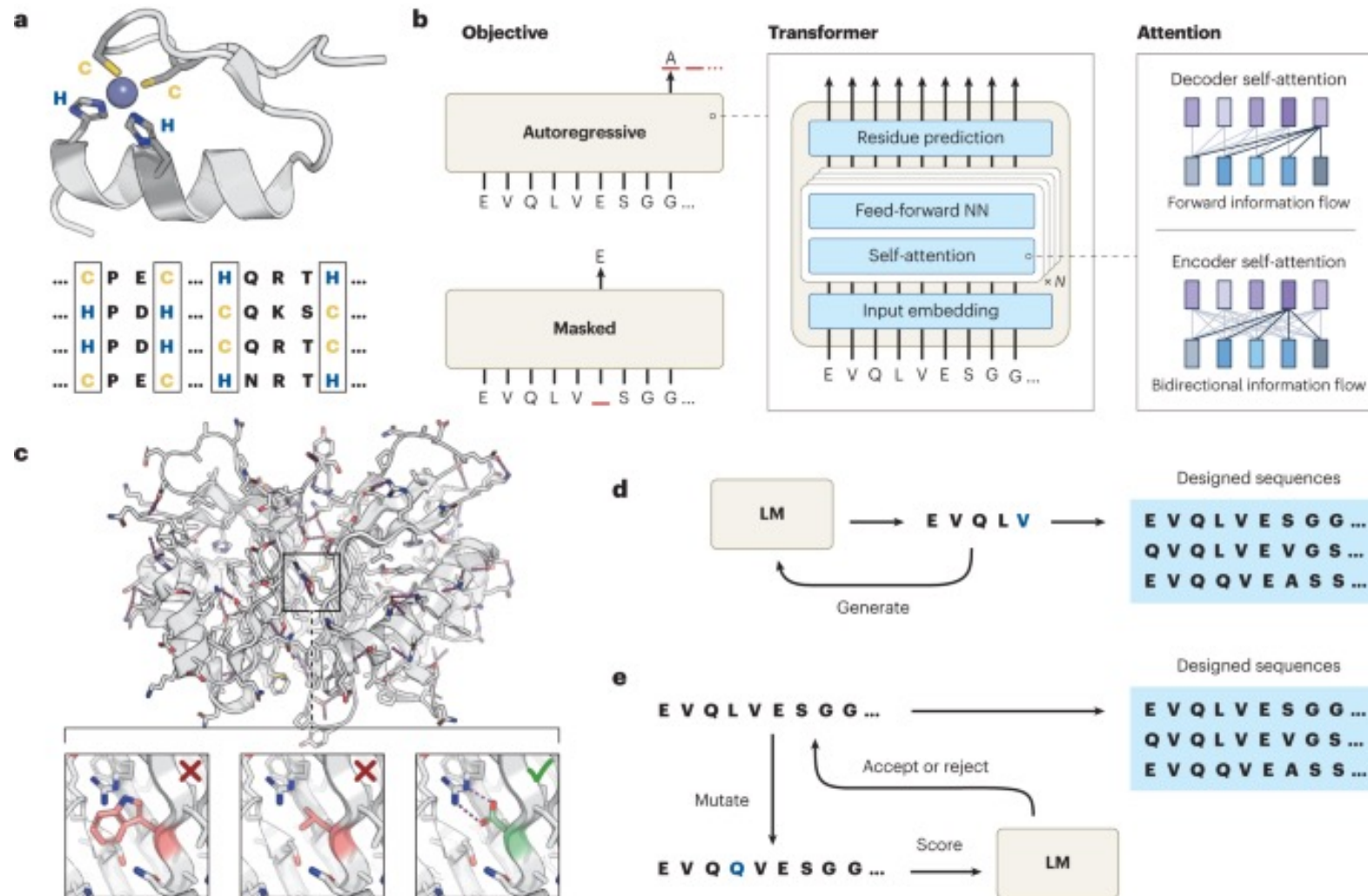


Figure source: Ruffolo and Madani. Designing proteins with language models, Nature Biotechnology, 2024.

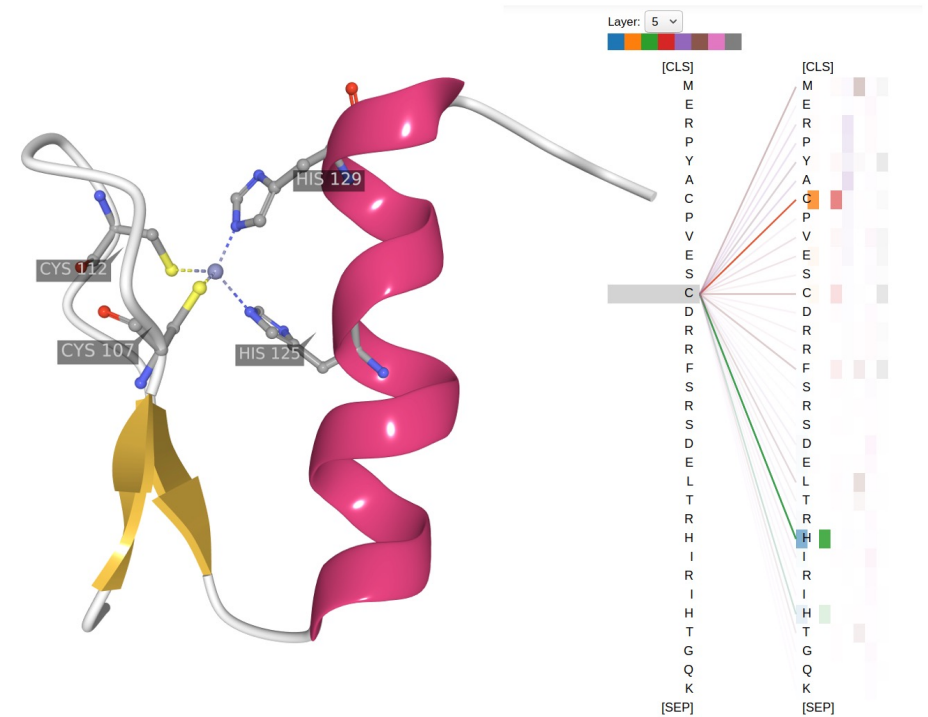
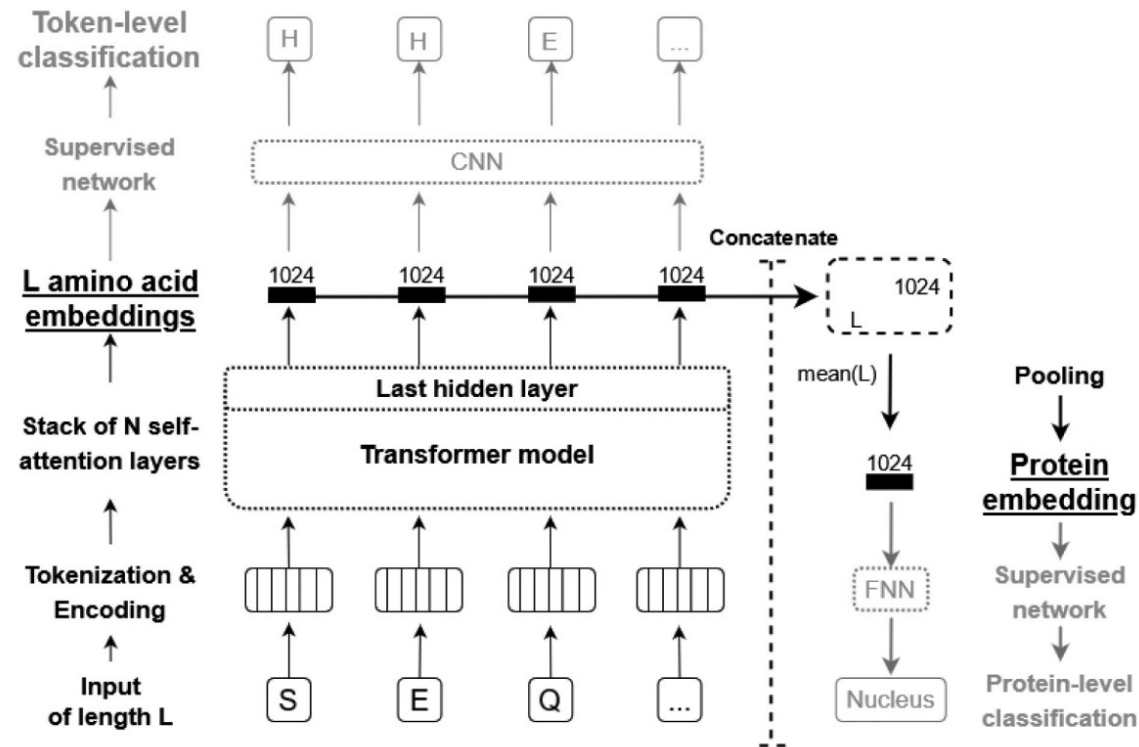
Relevant review/benchmark papers

Learning functional properties of proteins with language models. Unsal, Atas, Albayrak, Turhan, Acar & Doğan. 2022.

Fine-tuning protein language models boosts predictions across diverse tasks. Schmirler, Heinzinger & Rost. Nature Machine Intelligence. Nature Communications, 2024.

Designing proteins with language models. Ruffolo and Madani. Nature Biotechnology, 2024.

ProtTrans (ProtBert)

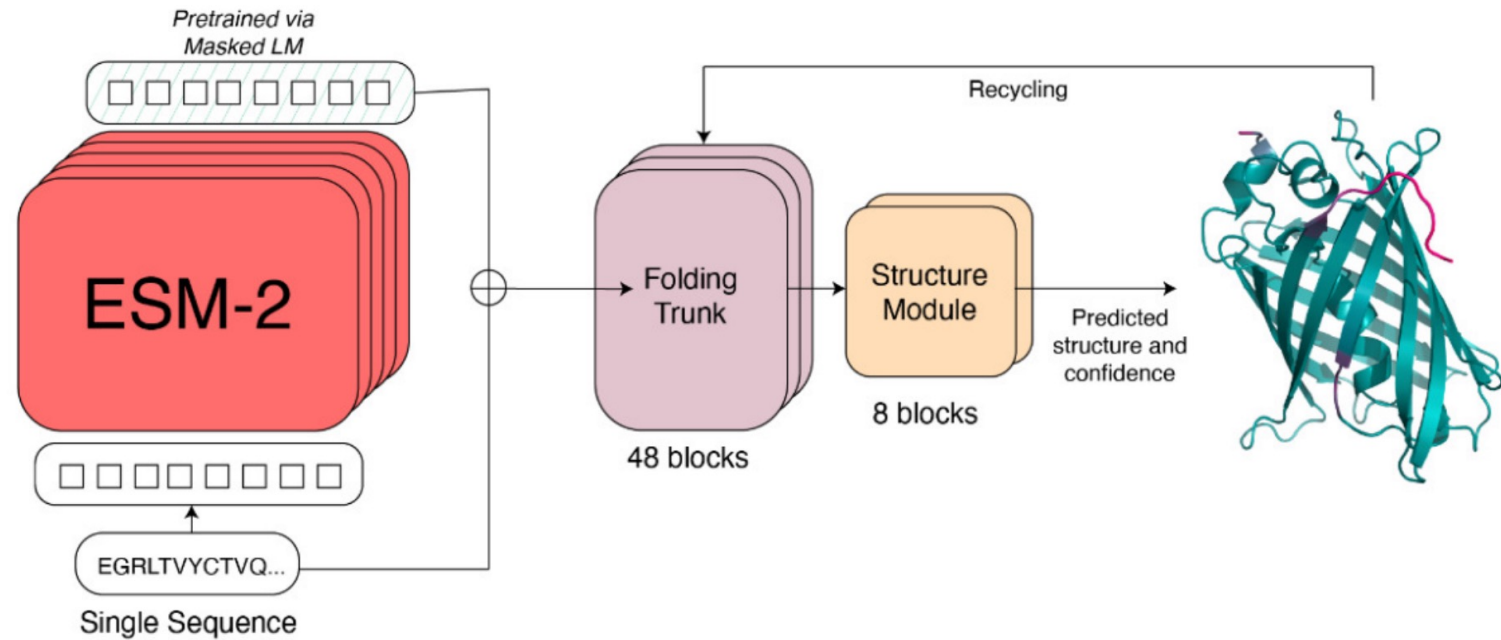


<https://github.com/agemagician/ProtTrans>

Bi-directional masked language modeling

- **Bidirectional** - the transformer model can read a sequence both forwards and backward
- **Masked language modeling loss** - pick random tokens in the input sequence and hide them (or mask them) from the model. Score the model based on whether or not it can guess this hidden token given the rest of the input sequence.

ESM2



- The ESM2 block is trained through **masked language modeling**.
- Does not use Multiple Sequence Alignment
- Trained across 250 million protein sequences spanning evolutionary diversity
- This objective enabled the model to learn inter and intra-dependencies

Evolutionary constraints

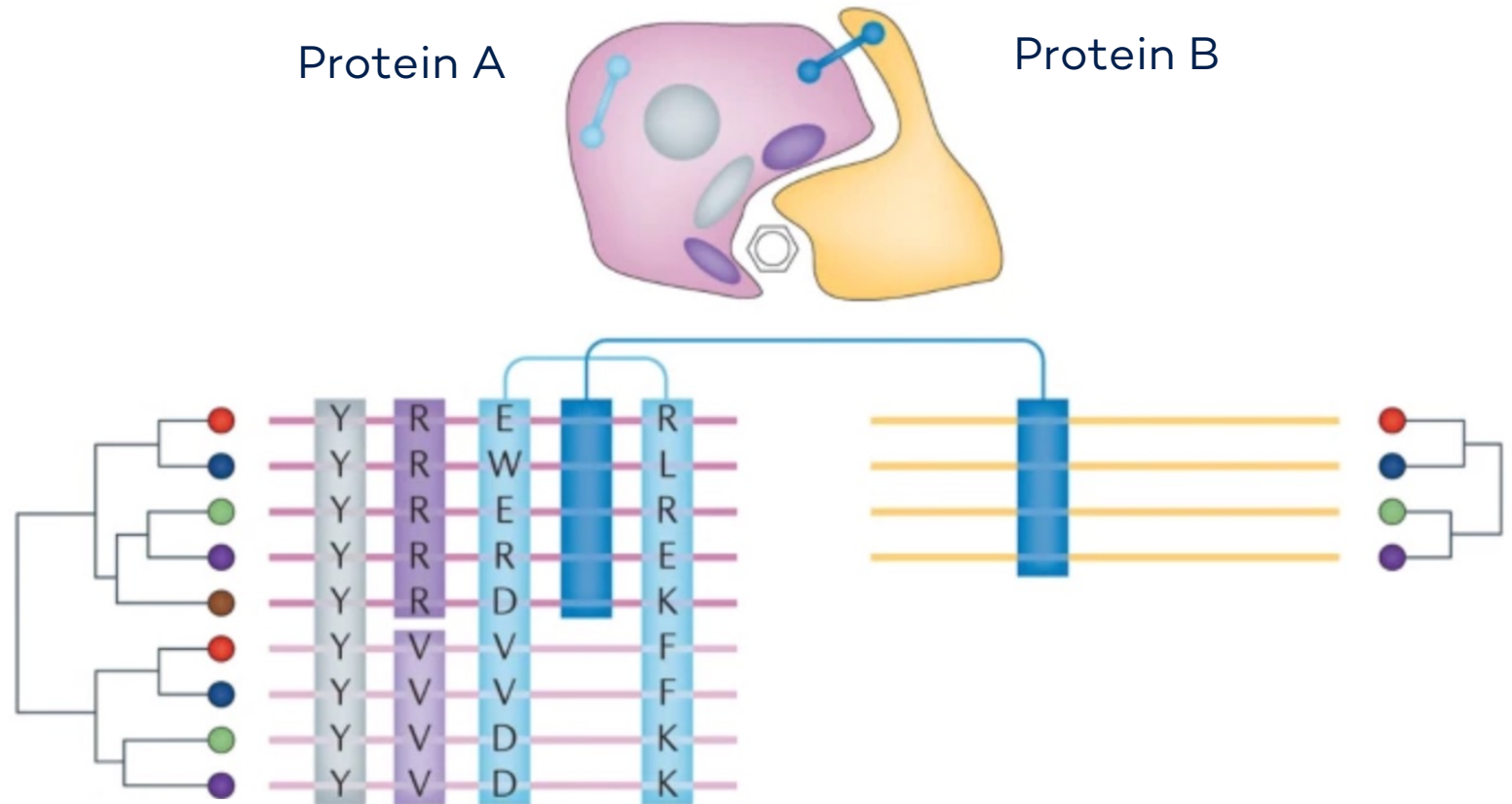
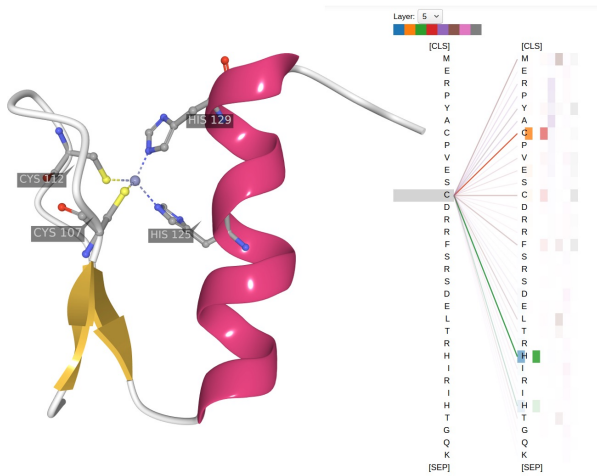
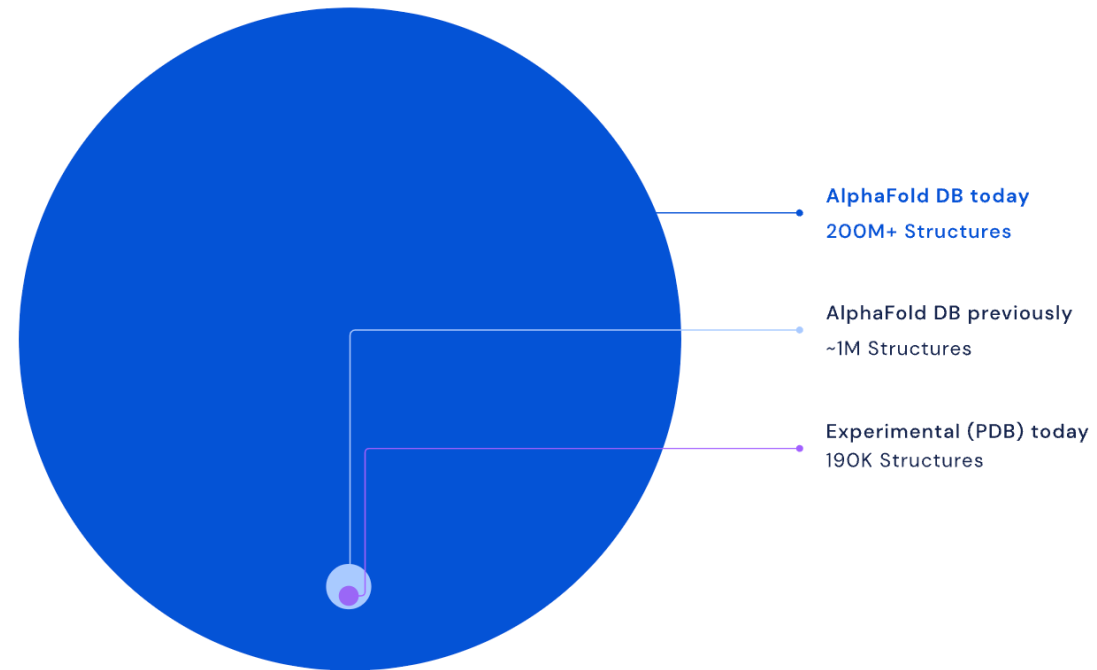
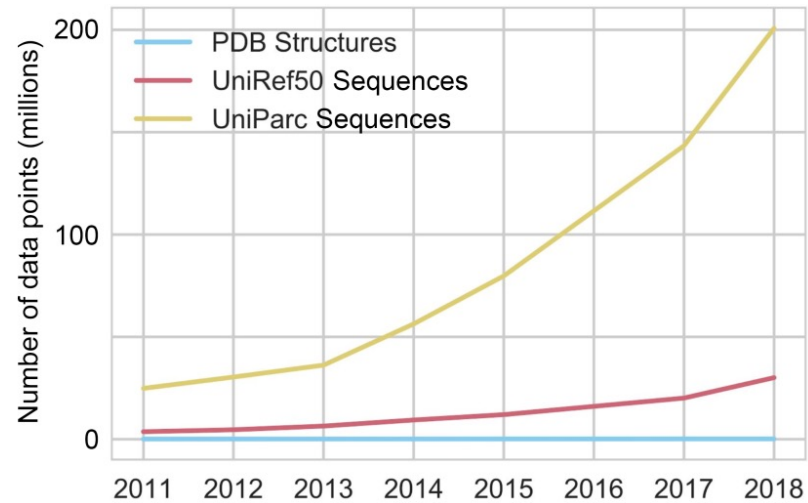


Figure adapted from: Juan et al. Emerging methods in protein co-evolution. Nature Reviews Genetics, 2013.

AlphaFold Revolution

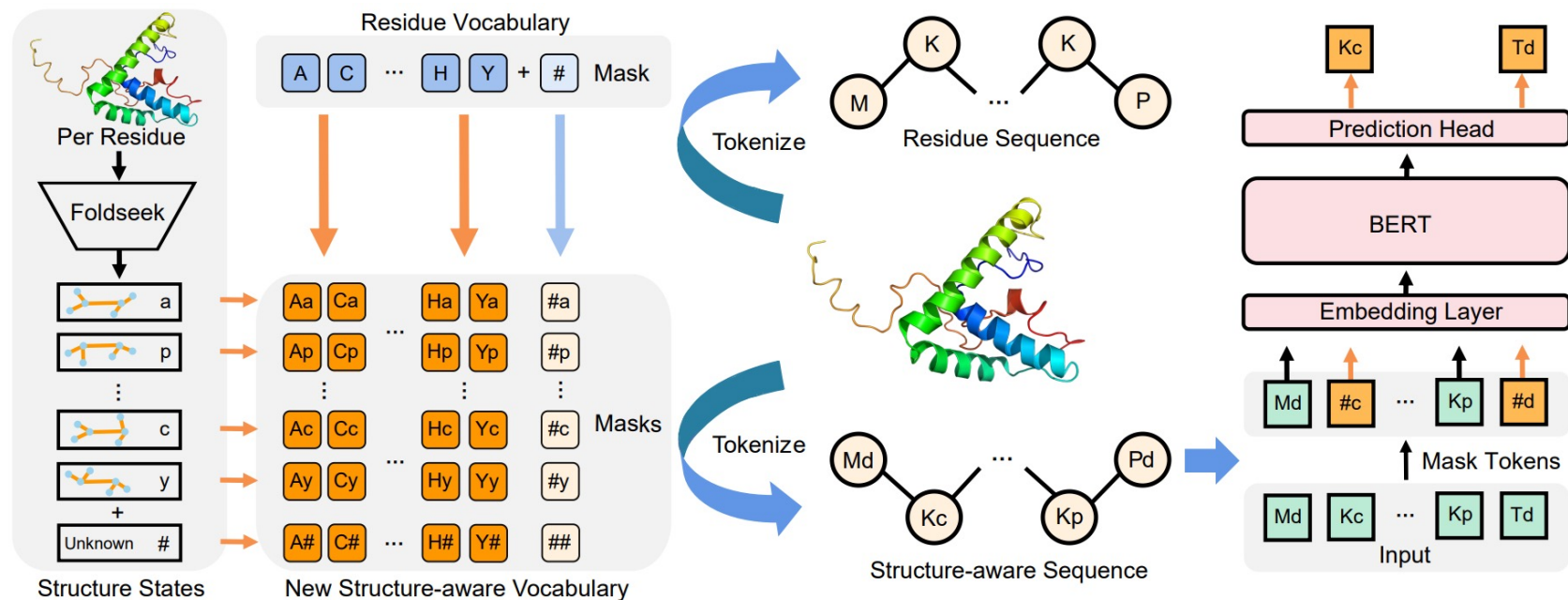


<https://moalquraishi.wordpress.com/2019/04/01/the-future-of-protein-science-will-not-be-supervised/>

<https://deepmind.google/discover/blog/alphafold-reveals-the-structure-of-the-protein-universe/>

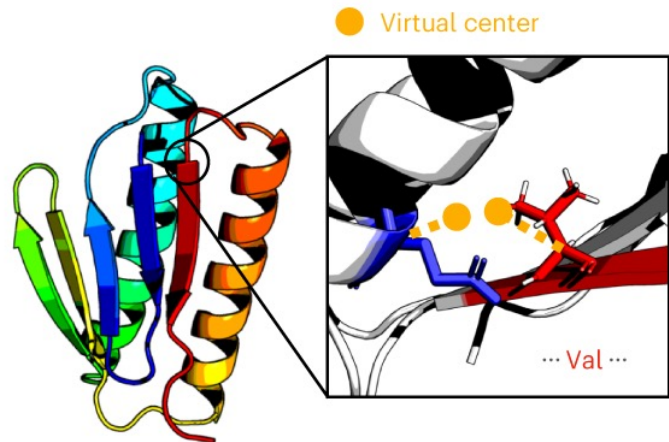
Structure informed PLMs

SaProt

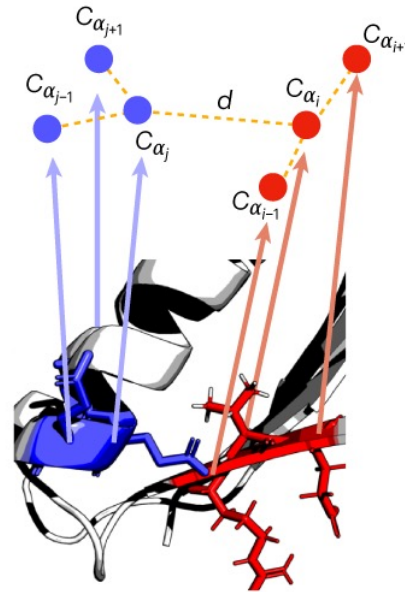


- Creates a structure-aware vocabulary.
- Each token encompasses both residue and structure information.
- SaProt employs the same network architecture and parameter size as the 650M version of ESM2.

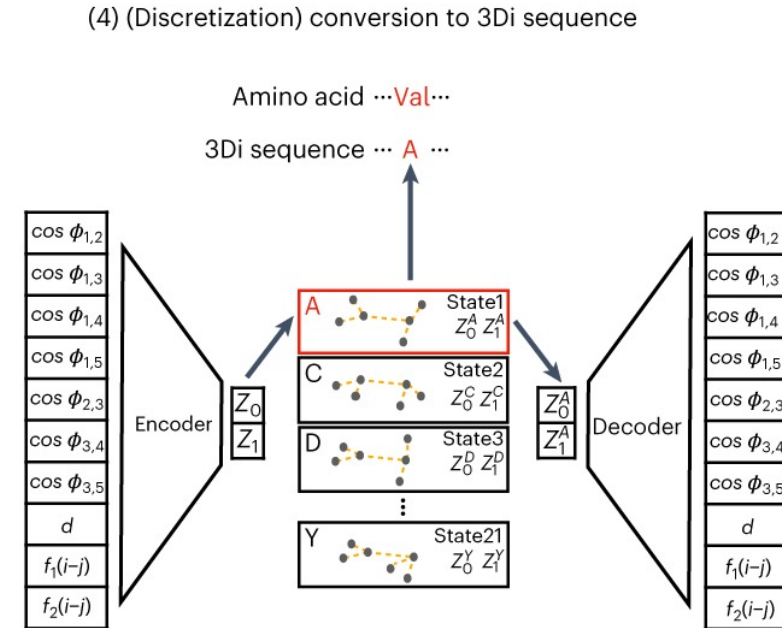
Foldseek – encoding of structures into 3Di sequences



(1) Find neighboring residues using virtual center



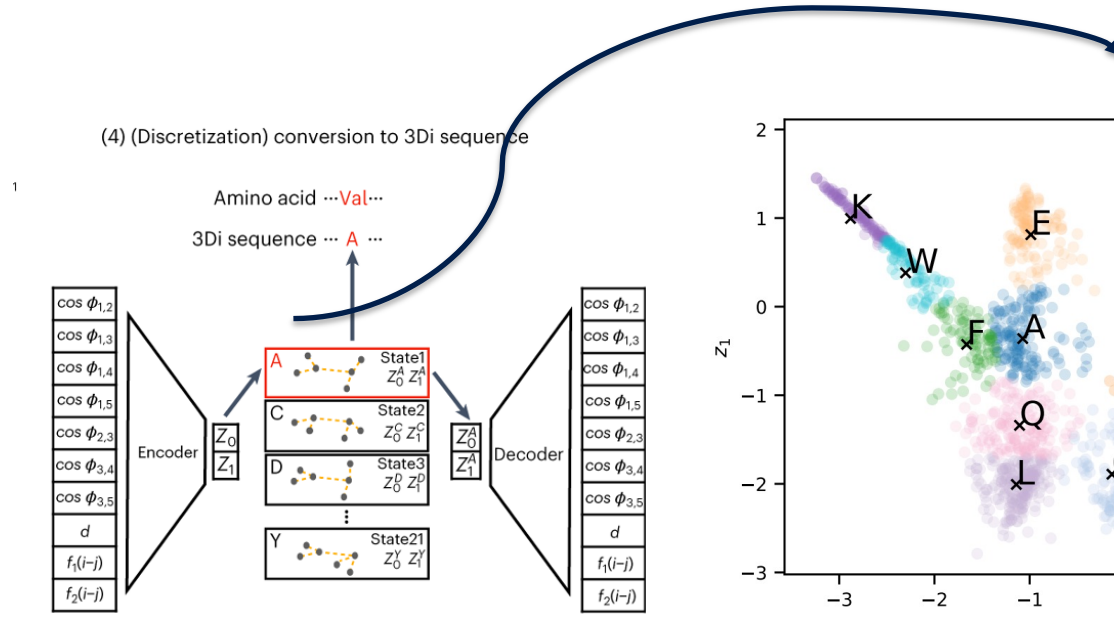
(2) Extract features



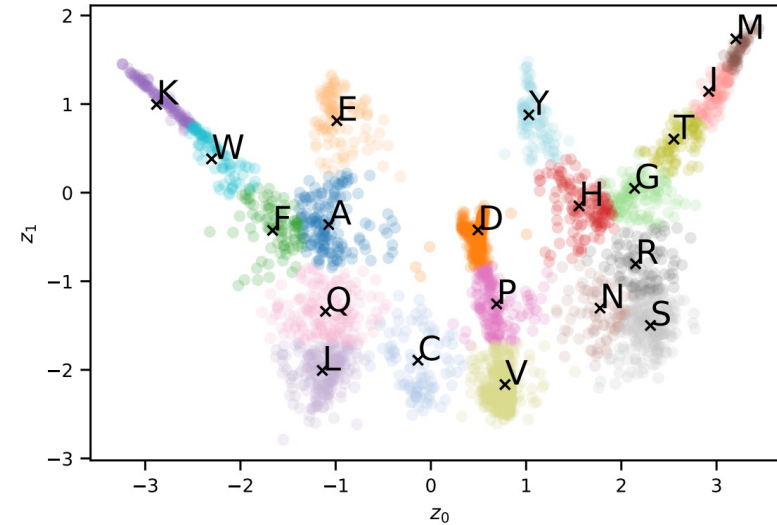
(3) Search 3Di state library

(4) (Training) predict features

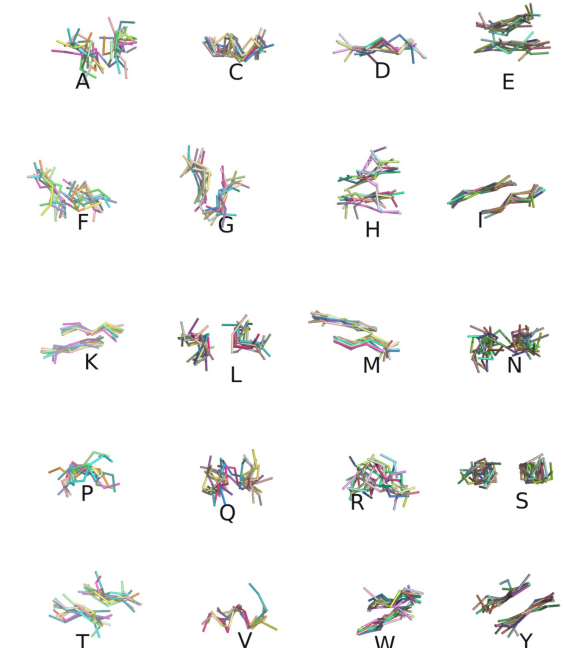
Foldseek – encoding of structures into 3Di sequences



Variational auto-encoder



Latent state representation
learned by the network



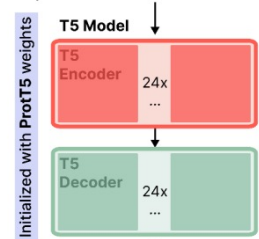
3Di state visualizations.

Structure informed PLMs

ProstT5

ProstT5 Architecture

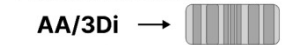
Input: AA or 3Di



Output: 3Di or AA

ProstT5 Applications

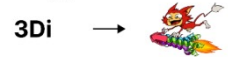
Foundation model



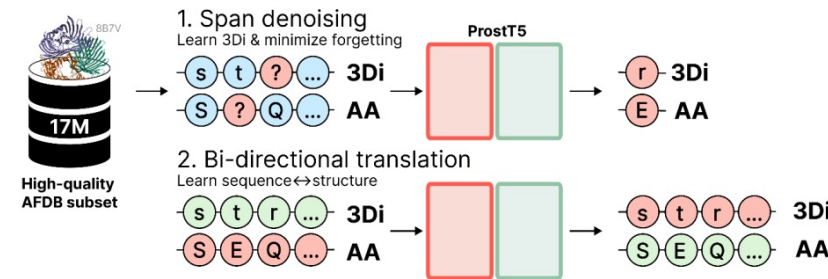
Inverse folding



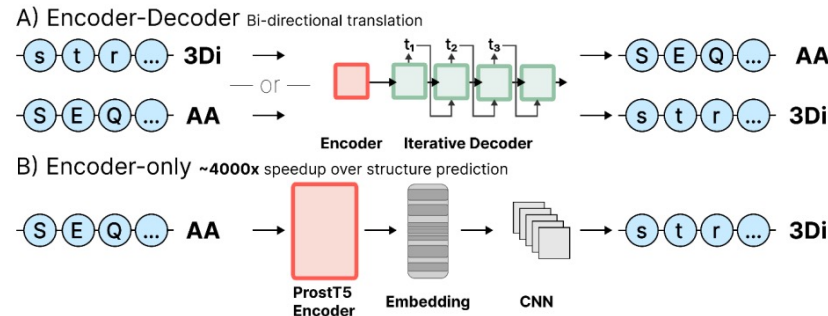
Sensitive structural
homology search



ProstT5 Pre-training



ProstT5 Inference



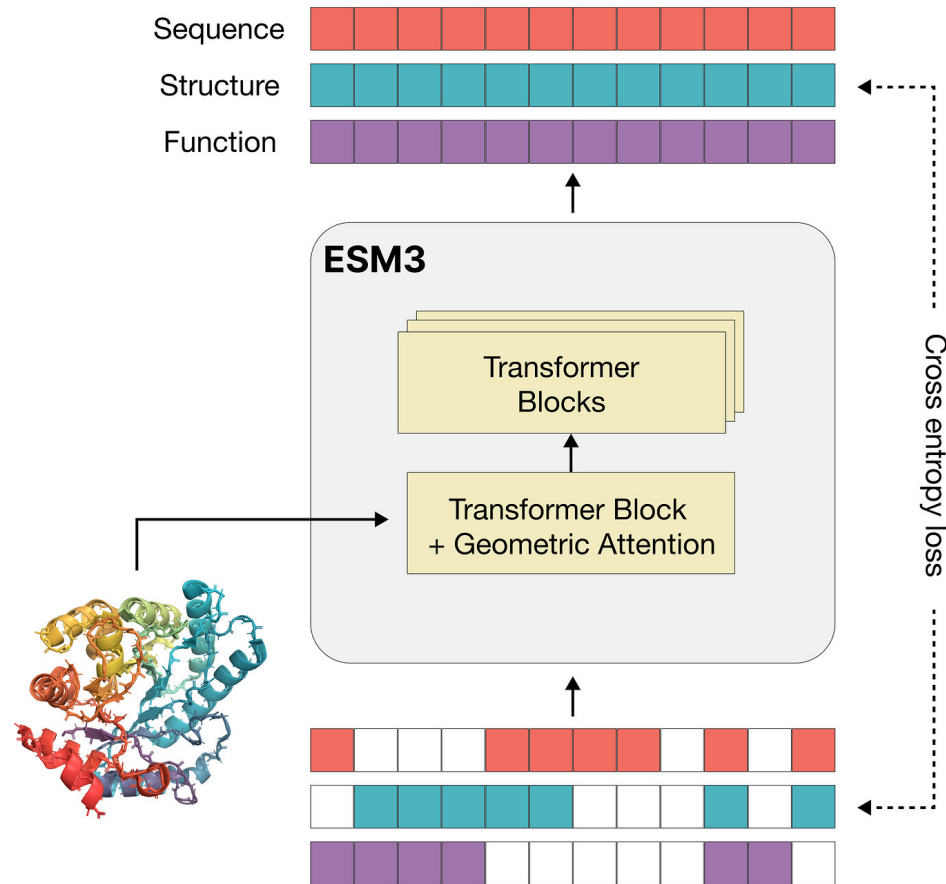
ProstT5 extends the ProtT5 model.

1. Applies the objective of ProtT5 (span-based denoising) to both, AAs and 3Di

2. Translates between aminoacid and structural sequence

Heinzinger et al. Bilingual Language Model for Protein Sequence and Structure. bioRxiv, 2024.

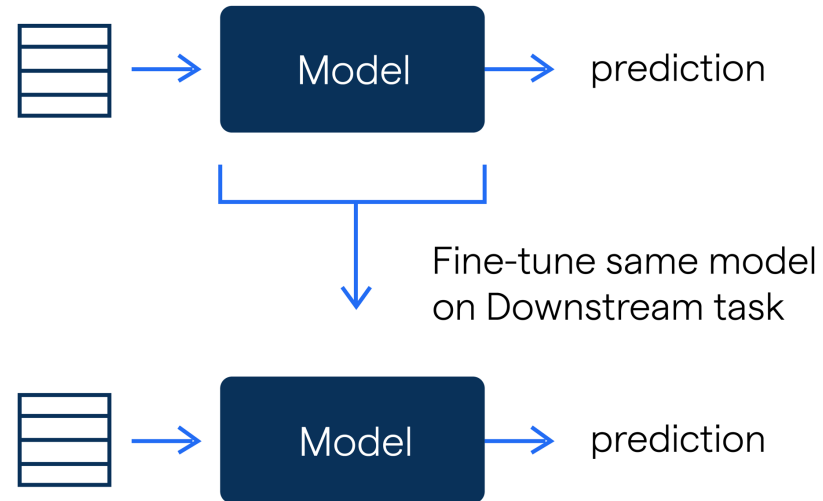
ESM3



ESM3 incorporates multiscale data into its training process.

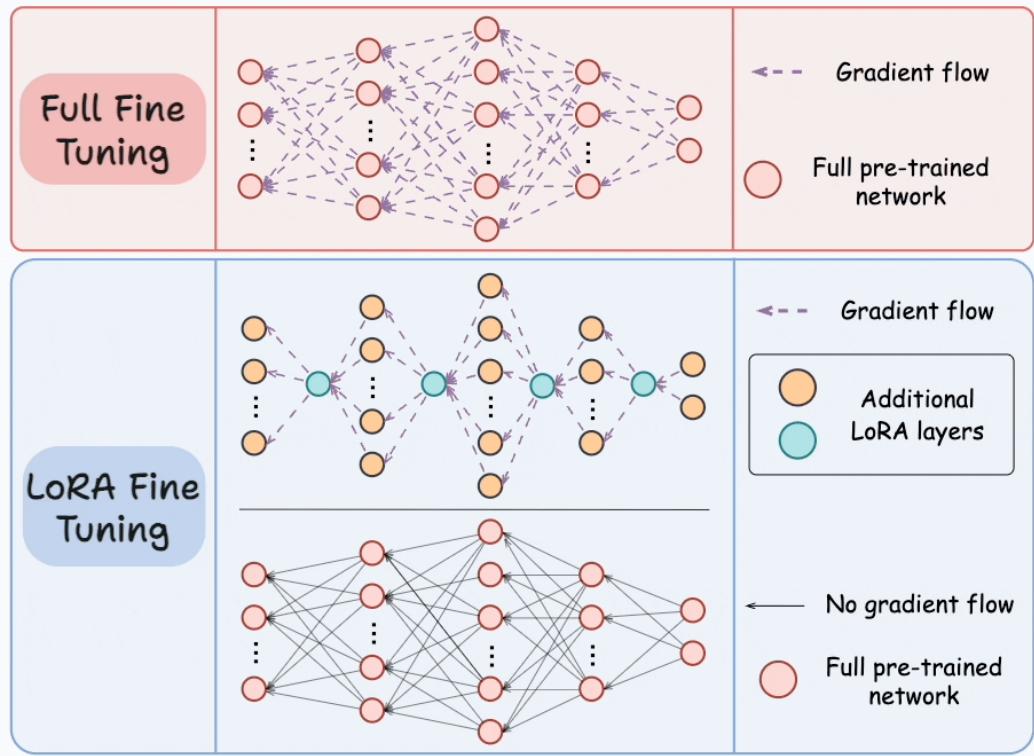
PART 3 – Finetuning with LORA

Fine-Tuning



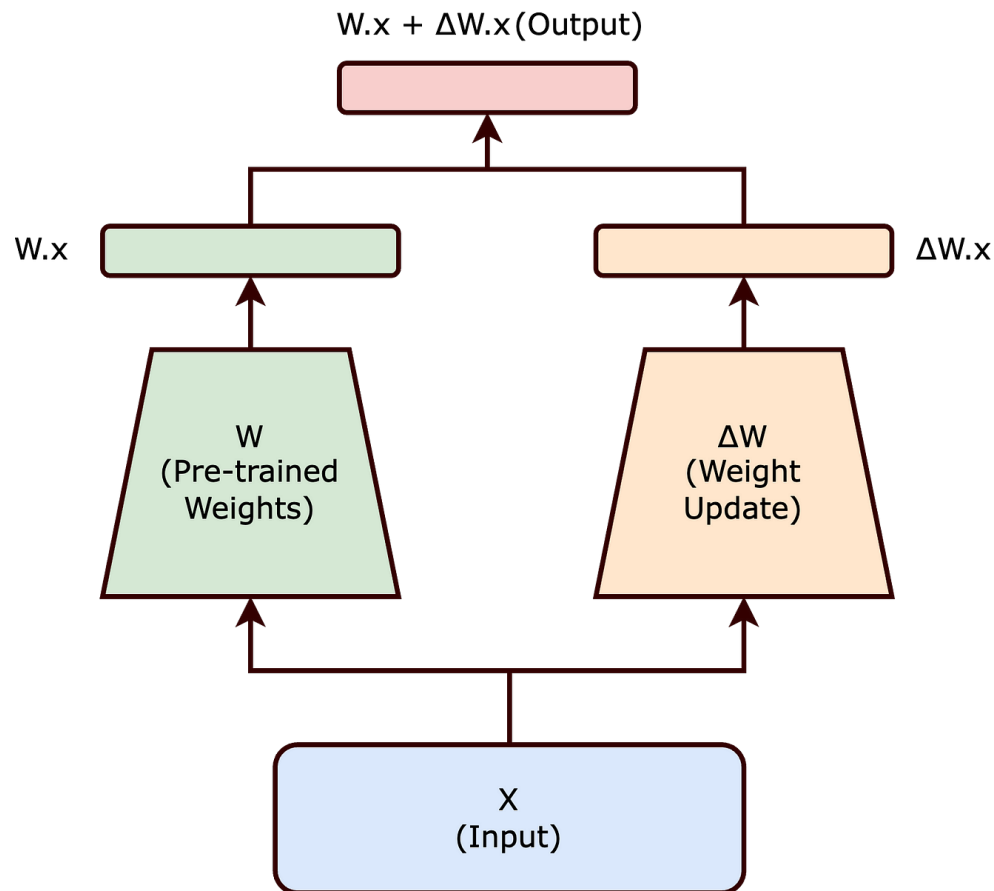
LORA – Low Rank Adaptation of LLMs

- Finetuning the model requires to store as many parameters as the original model
- LORA enables task specific parameter increment encoded by a much smaller size set of parameters

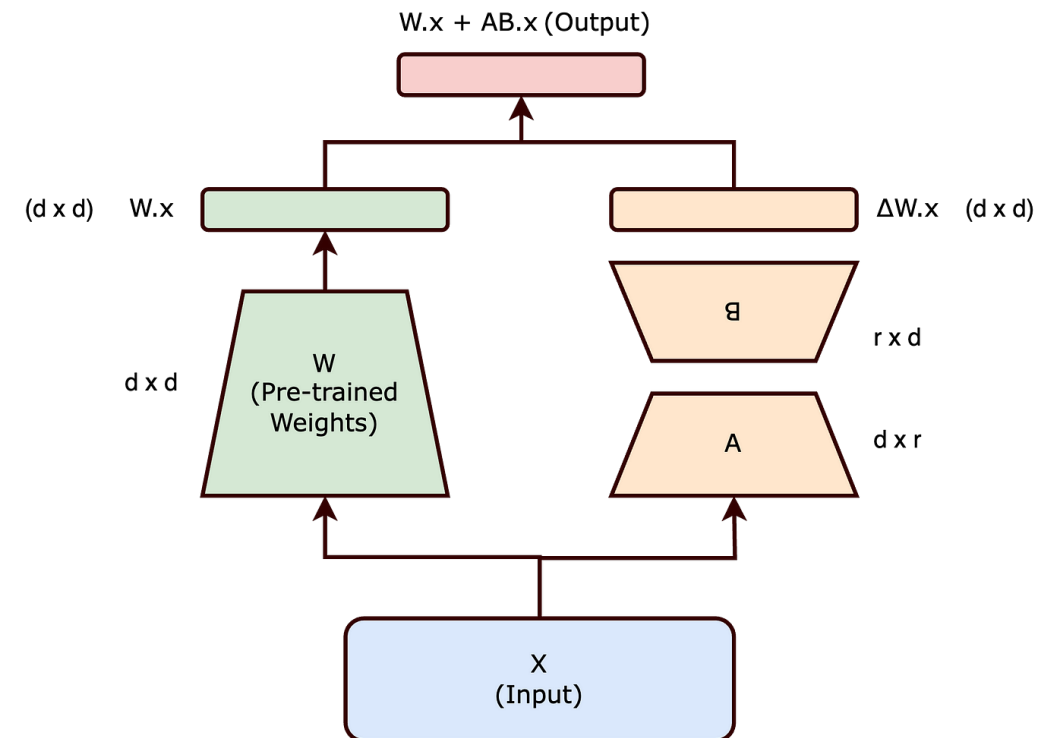


Traditional fine tuning vs LORA

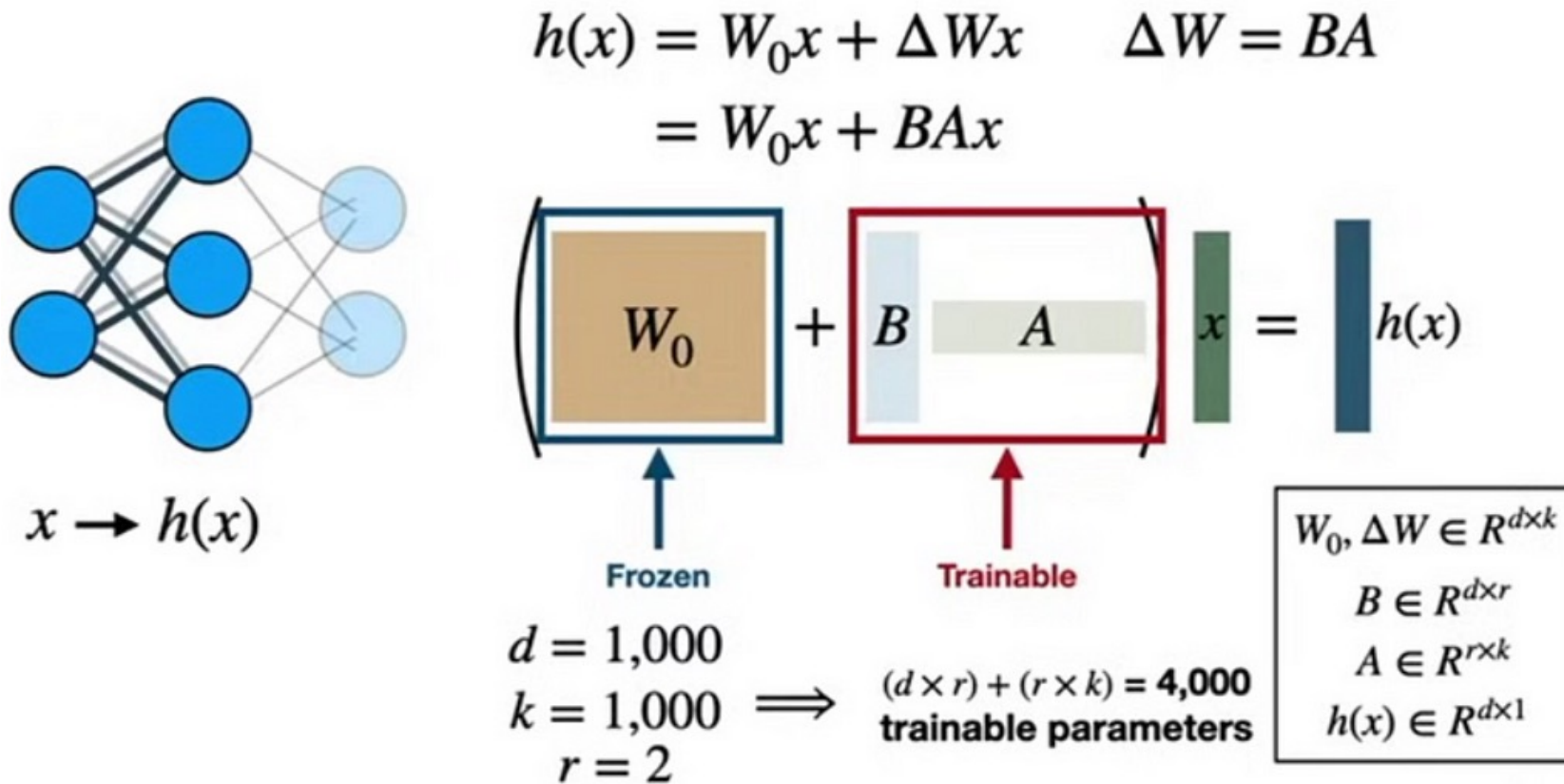
Finetuning



LORA



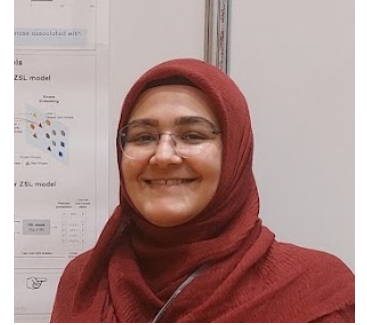
LORA – Low Rank Adaptation of LLMs



PART 4 –The task in the demo today

Demo today we will work on today

- Generating embeddings using two protein language models
- Fine tuning ESM for the downstream task
- Fine tuning with LoRA

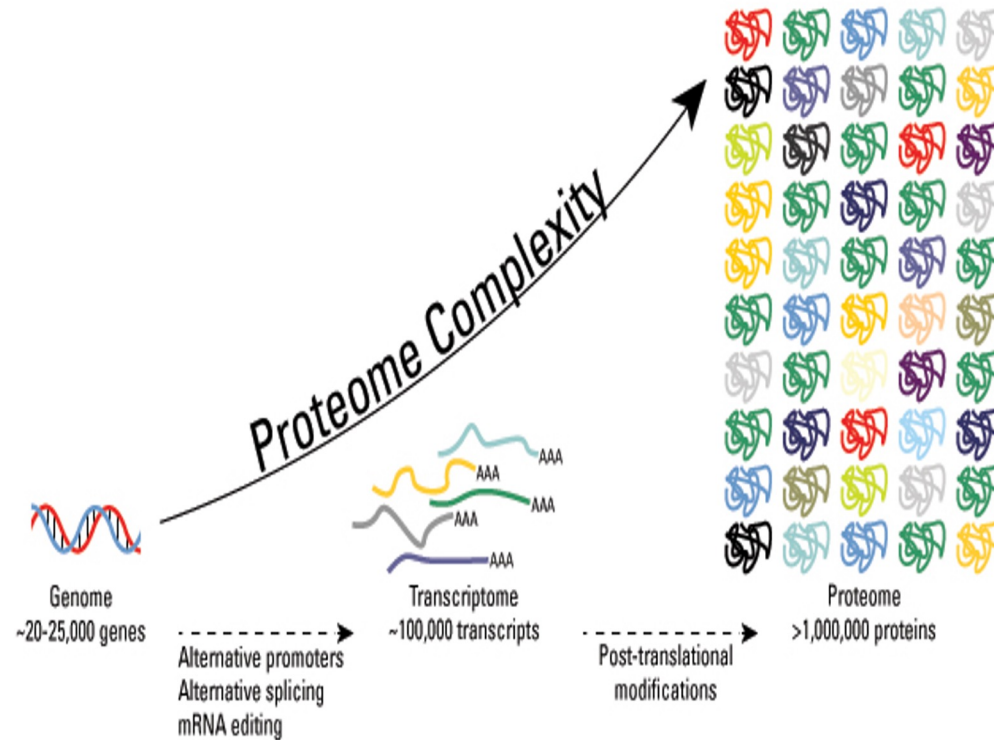


Zeynep Işık

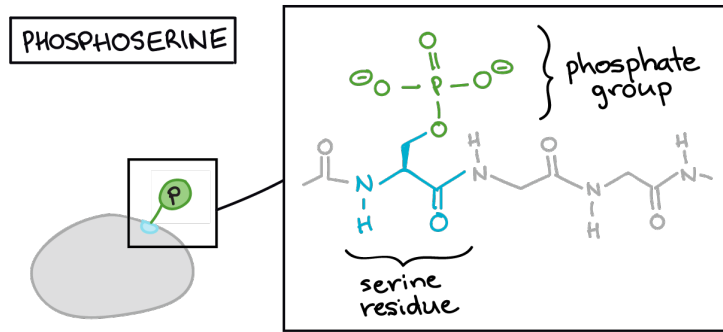


Mert Pekey

Post-translational Modifications



Phosphorylation



Phosphorylation

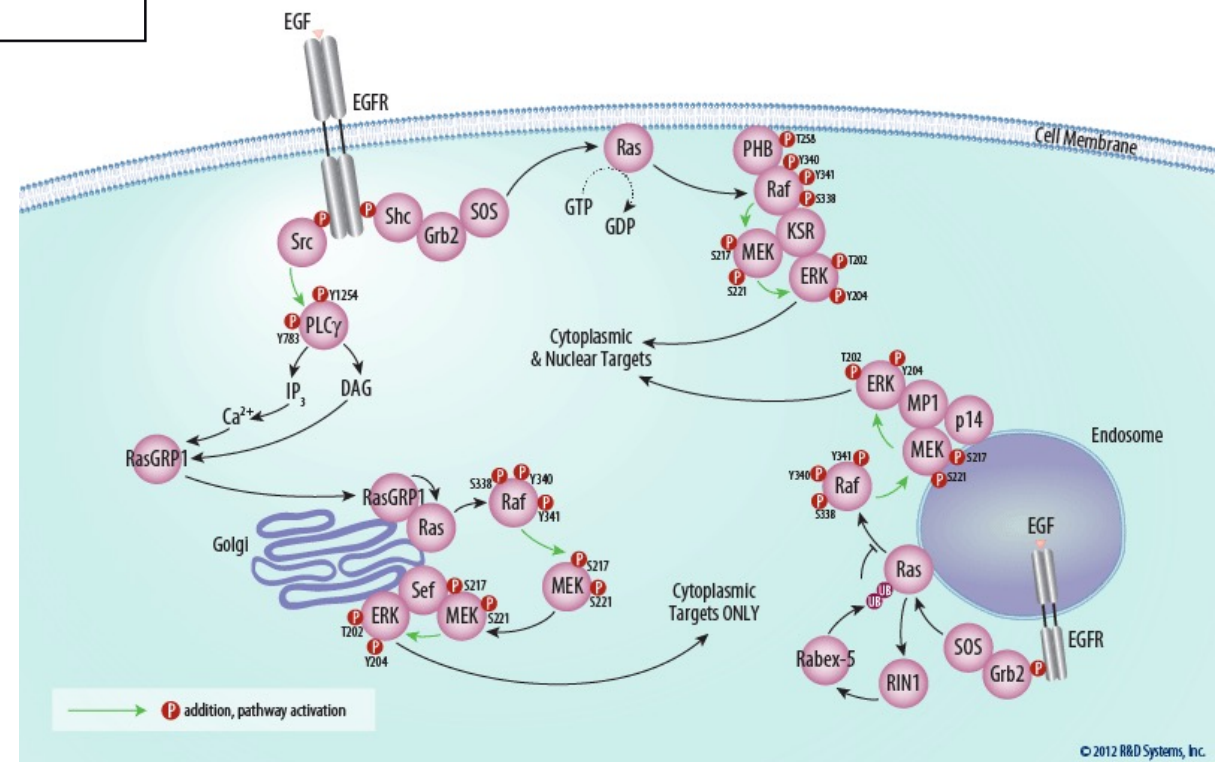
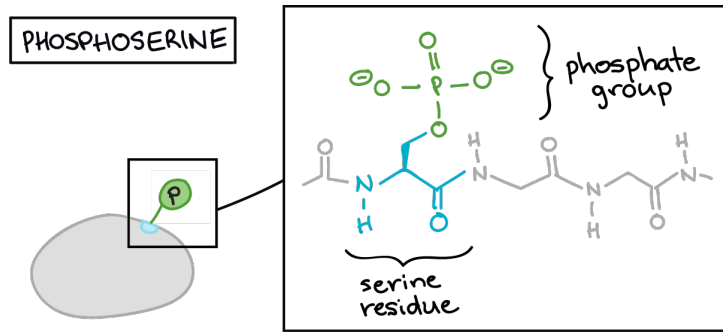
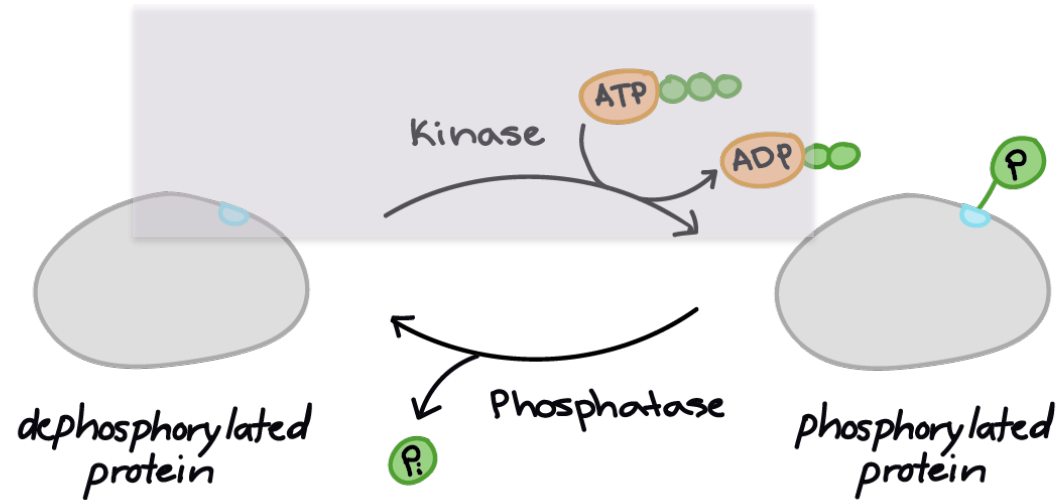


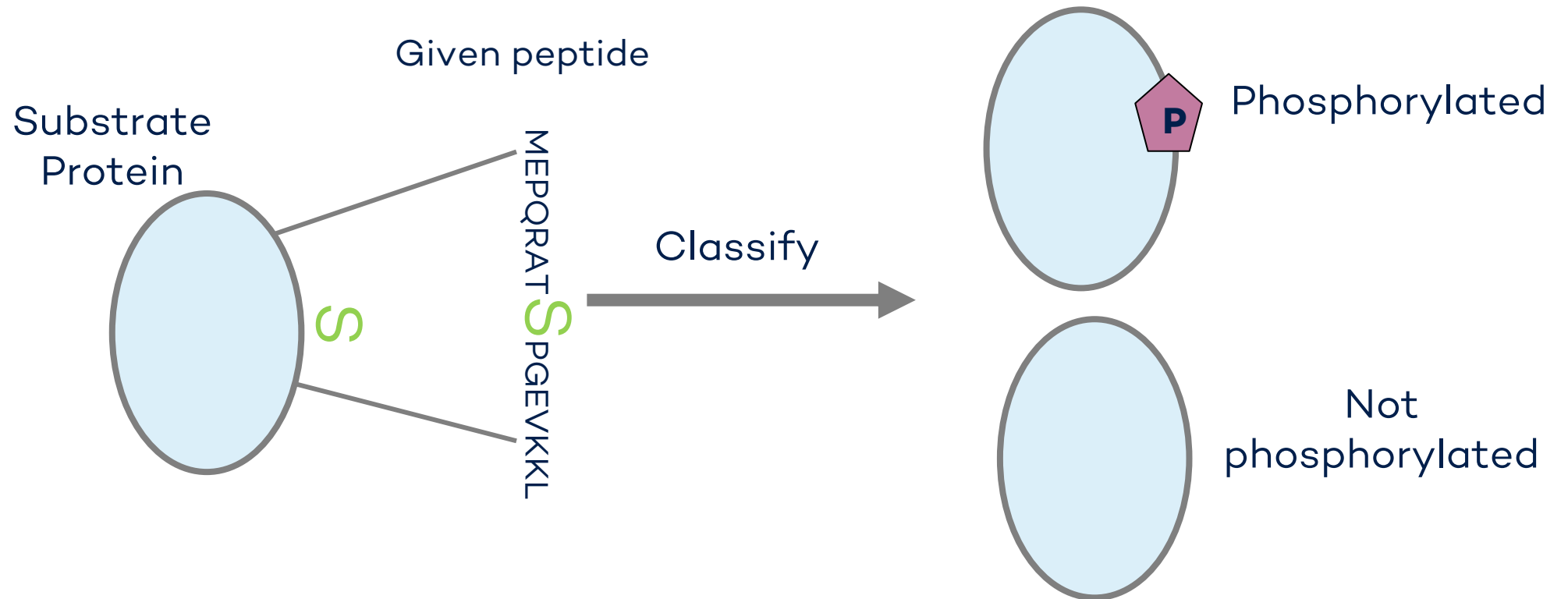
Figure source: <https://www.rndsystems.com/resources/articles/erk-signal-transduction-pathway>

Kinases



Supervised Binary Classification Task

Given a phosphosite sequence, **predict** whether the site will be phosphorylated or not



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Thank you!

